# Manufacturing and Service Operations
# MSO 40.012

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# Summary Course 2

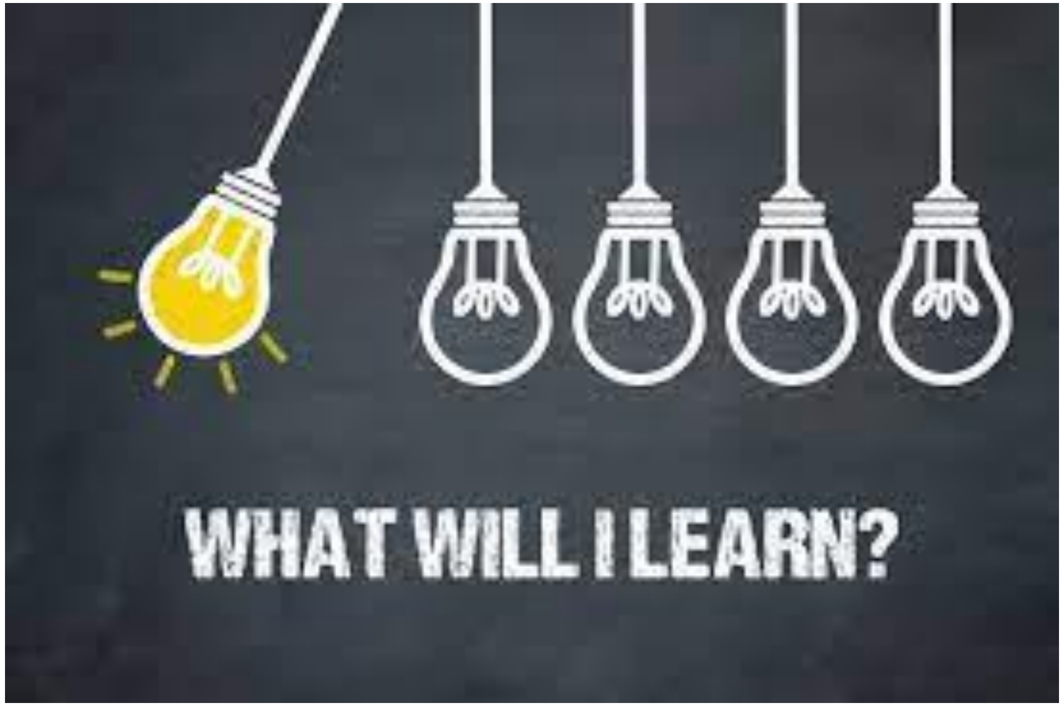
- **Forecasting Necessity**: While forecasts are never 100% accurate, they are essential for planning resources, aligning with suppliers, and satisfying customers to avoid losing market share.
- **Impact of Inaccuracy**: Poor forecasting leads to value leakage, such as lost revenue from out-of-stocks or increased costs from excess inventory and emergency logistics.
- **Forecast Characteristics**: Accuracy generally improves when data is aggregated (by product family rather than SKU) and when the planning horizon is short-term.
- **Seven-Step Forecasting Process**: A structured approach involves determining the use, selecting items, setting horizons, choosing models, gathering data, forecasting, and validating results.
- **Qualitative vs. Quantitative Methods**: Qualitative methods like the Delphi method use expert consensus, while quantitative methods (Time Series, Regression) use historical data for objective decision support.
- **Key Performance Indicators (KPIs)**: Success is measured using Forecast Accuracy, Forecast Bias (to detect over/under-forecasting), and Forecast Consistency (month-on-month fluctuation).
- **Planning Time Fences (PTF)**: Effective planning requires a "frozen fence" (typically 4 weeks) where changes are restricted, followed by a PTF (approx. 3 months) for the Master Production Schedule.
- **Aggregate Planning Goals**: The objective is to determine production rates, workforce size, and inventory levels to maximize profit while meeting demand over a fixed horizon.

# Agenda Course 3

- MPS or Fixed planning horizon problem
  - Introduction
  - Chase vs level production strategy
  - Simple Heuristics: lot for lot/fixed EOQ
  - Specialized heuristics

- MRP
  - BOM
  - MRP
  - Introduction to ERP

# Learning Objectives




**Understand** how MPS differs from aggregate planning

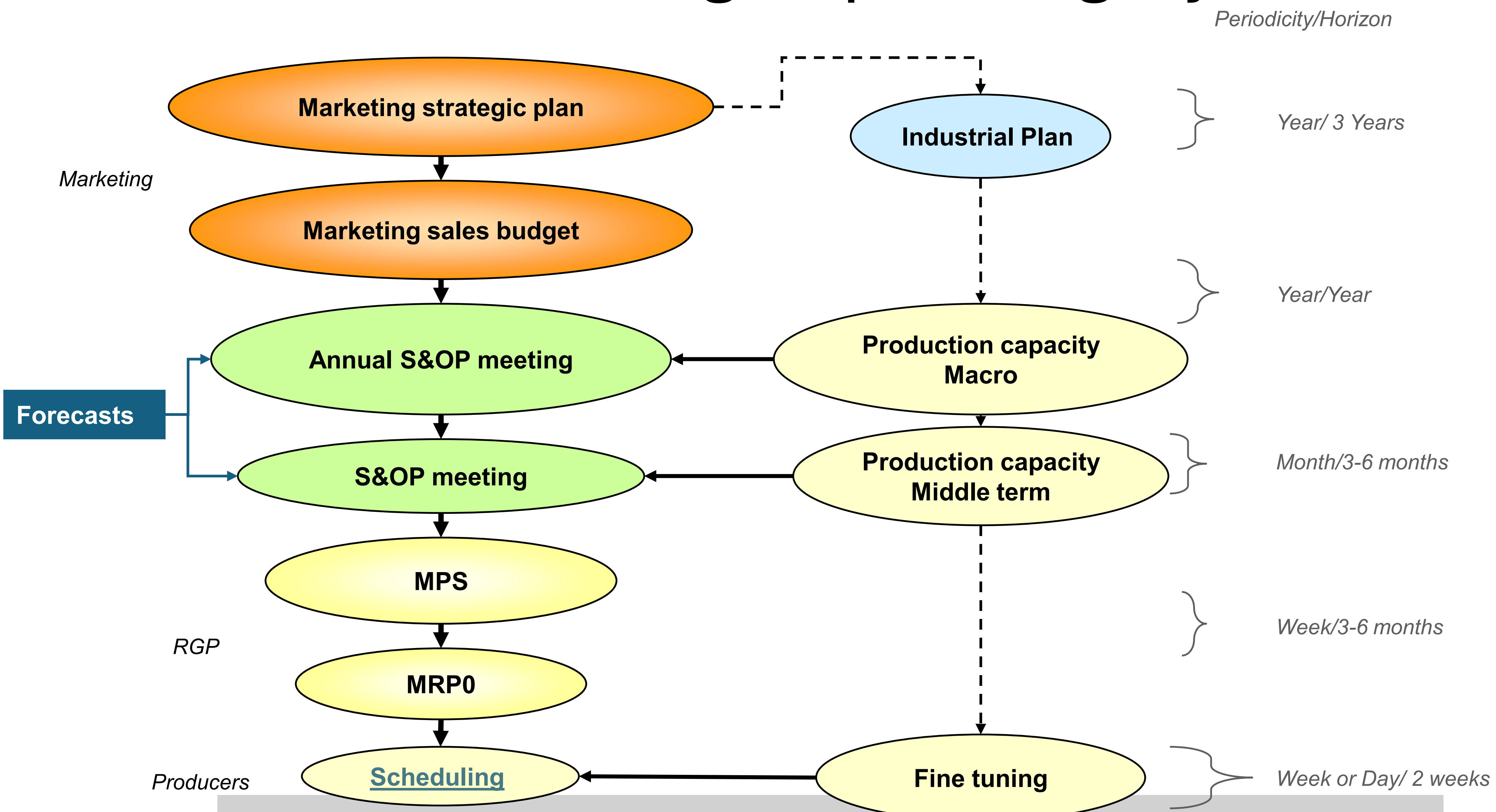
**Describe** the 2 approaches to production planning

**Develop** a production plan

**Compute** different production plans based on various models

**Understand** the different costs attached to each model

# Where are we among all planning layers?



# The MPS: Master Production Schedule

- Specifies what is to be made, how much and when
- Must be in accordance with the aggregate production plan
- Aggregate production plan sets the overall level of output in broad terms; MPS drills down to SKU
- As the process moves from planning to execution, each step must be tested for feasibility
- The MPS is the result of the production planning process

# The MPS: Master Production Schedule

MPS is established in terms of specific products

Schedule must be followed for a reasonable length of time

The MPS is quite often fixed or frozen in the near term part of the plan

The MPS is a rolling schedule

The MPS is a statement of what is to be produced, not a forecast of demand

# From aggregate plan to MPS

From product family to product

**Aggregate Plan**
(Shows the total quantity of amplifiers)

| Months | January | February |
|---|---|---|
| | 1,500 | 1,200 |

**Master Production Schedule**
(Shows the specific type and quantity of amplifier to be produced)

| Weeks | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 240-watt amplifier | 100 | | 100 | | 100 | | 100 | |
| 150-watt amplifier | | 500 | | 500 | | 450 | | 450 |
| 75-watt amplifier | | | 300 | | | | 100 | |

Figure 14.2

**The Aggregate Plan Is the Basis for Development of the Master Production Schedule**

# From aggregate plan to MPS

From product family to product:

- Fixed ratio based on historical data applied to the aggregate forecast
    - 20% for PFa
    - 50% for PFb
    - 30% for PFc

- Customer orders on hand and customer operative forecast

- Safety stock requirements

# Chase and level: 2 opposite approaches

Given: Current inventory level

Given: Forecast of demand by period

| Time Period | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| *Forecast* | | 10 | 15 | 20 | 22 | 25 | 28 |
| *Production Plan* | | | | | | | |
| *Projected Available Balance* | 30 | | | | | | |

Entered: Production plan to be made during that period

Calculated: Expected inventory a the END of period 1

$$I_t = I_{t-1} + P_t - F_t$$

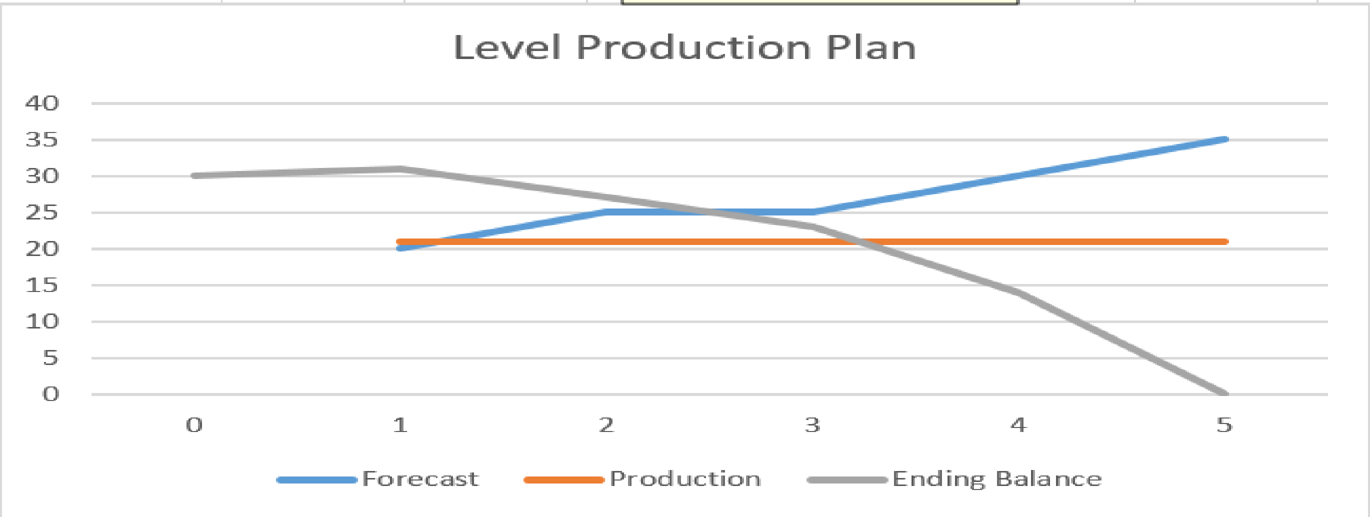
2 main strategies to craft the production plan:

- LEVEL strategy
- CHASE strategy

# Level: keep production rate stable

| Time Period | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Forecast | | 20 | 25 | 25 | 30 | 35 |
| Production | | 21 | 21 | 21 | 21 | 21 |
| Ending Balance | 30 | 31 | 27 | 23 | 14 | 0 |

**Roxane Desmicht:**
Inventory on hand



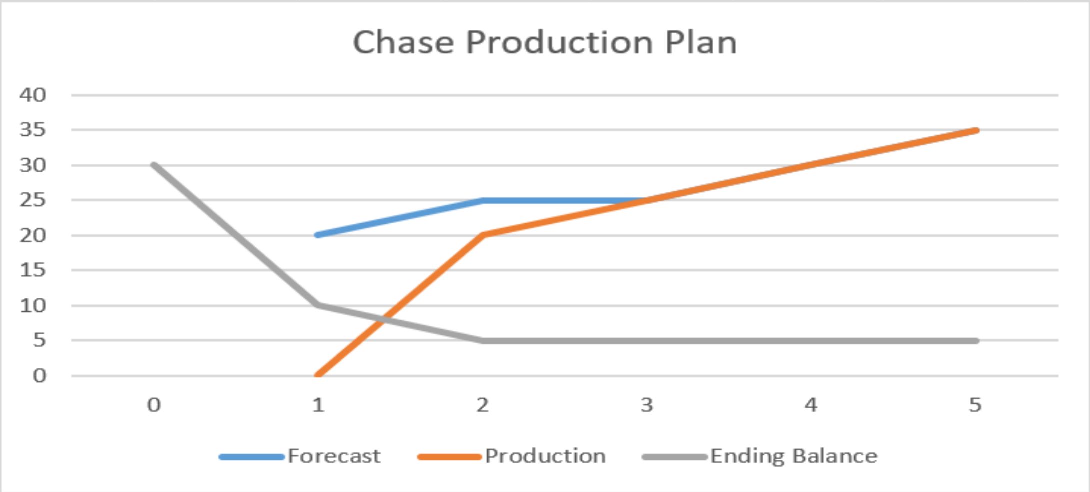

- PROS:
  - Smooth plan for production with predictable resources requirements
  - Minimize need for short-term overtime
  - Minimize changes and switching costs
- CONS
  - It could result in shortages during peak demand period
  - It could result in high inventory during low demand period

# Chase: match production to the demand

| Time Period | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Forecast | | 20 | 25 | 25 | 30 | 35 |
| Production | | 0 | 20 | 25 | 30 | 35 |
| Ending Balance | 30 | 10 | 5 | 5 | 5 | 5 |

**Roxane Desmicht:**
Inventory on hand

**Roxane Desmicht:**
Desired ending inventory of 5




- PROS:
  - Minimizes obsolescence risk
  - Minimize inventory
- CONS
  - Bring swings to production with the need to readjust resources in each period
  - Create a poor equipment utilization at times
  - Might still end up in shortages if production cannot fulfil a major upswing

# Level or Chase: more hybrid in reality

- Leverage on outsourcing to absorb peaks

- Explore cost trade-offs:
  - Lot sizing
  - Fixed production horizon problem

# Fixed Planning Horizon Problem

Find a production plan
Which optimize costs
We have 2 competing costs:
Setup costs
Inventory cost

- Demand
  - Constant vs **Variable**
  - **Known** vs Random
  - **Continuous** vs Discrete
- Lead time
  - **Instantaneous**
  - Constant or Variable (deterministic/ stochastic)
- Dependence of items
  - **Independent**
  - Correlated
  - Indentured
- Review Time
  - **Continuous**
  - Periodic
- Number of Echelons
  - **One**
  - Multi (>1)
- Capacity / Resources
  - **Unlimited**
  - Limited / Constrained

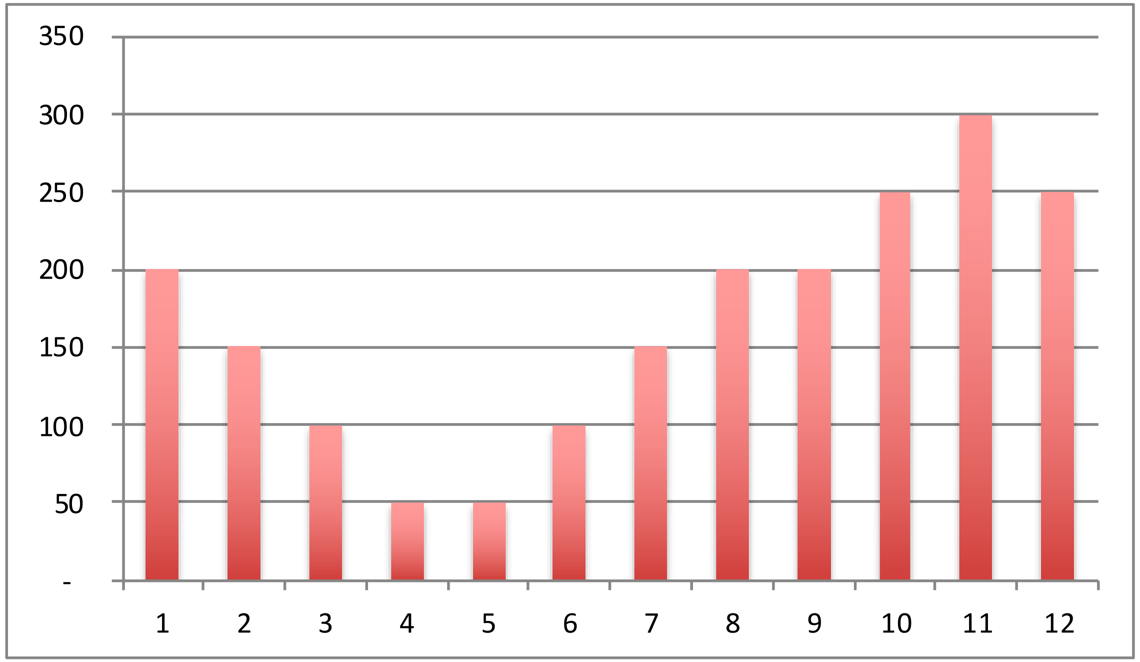
- Discounts
  - **None**
  - All Units or Incremental
- Excess Demand
  - **None**
  - All orders are backordered
  - Lost orders
  - Substitution
- Perishability
  - **None**
  - Uniform with time
- Planning Horizon
  - Single Period
  - **Finite Period**
  - Infinite
- Number of Items
  - **One**
  - Many
- Form of Product
  - **Single Stage**
  - Multi-Stage

# Fixed Planning Horizon Problem through an example

Katong &Co has signed a year contract with Astar to provide its Zigbee modules. It is a 1 year contract for 2000 pieces with below demand profile. The demand is known and fixed for a specific product over the overall horizon.

Katong &Co needs to decide the best production approach to fulfill the demand.

| Month | Demand |
|---|---|
| 1 | 200 |
| 2 | 150 |
| 3 | 100 |
| 4 | 50 |
| 5 | 50 |
| 6 | 100 |
| 7 | 150 |
| 8 | 200 |
| 9 | 200 |
| 10 | 250 |
| 11 | 300 |
| 12 | 250 |
| Total: | 2 000 |



# Our constraints

**Cost:**

| | |
|---|---|
| Production set-up cost/run: | 500 SGD |
| Unit cost: | 50 SGD |
| Holding cost: | 1SGD/unit/month |

**Other:**

Modules made in a period are ready to use in that period.
There are no holding costs for modules made and shipped in the same month, holding charges apply if the module is delivered in subsequent months.

**Objective**: find the lowest production cost for this contract.

# Methods to solve the problem

Simple heuristics:
- One time run
- Lot for Lot
- EOQ

Specialized heuristics:
- Silver Meal
- ~~Least unit cost~~

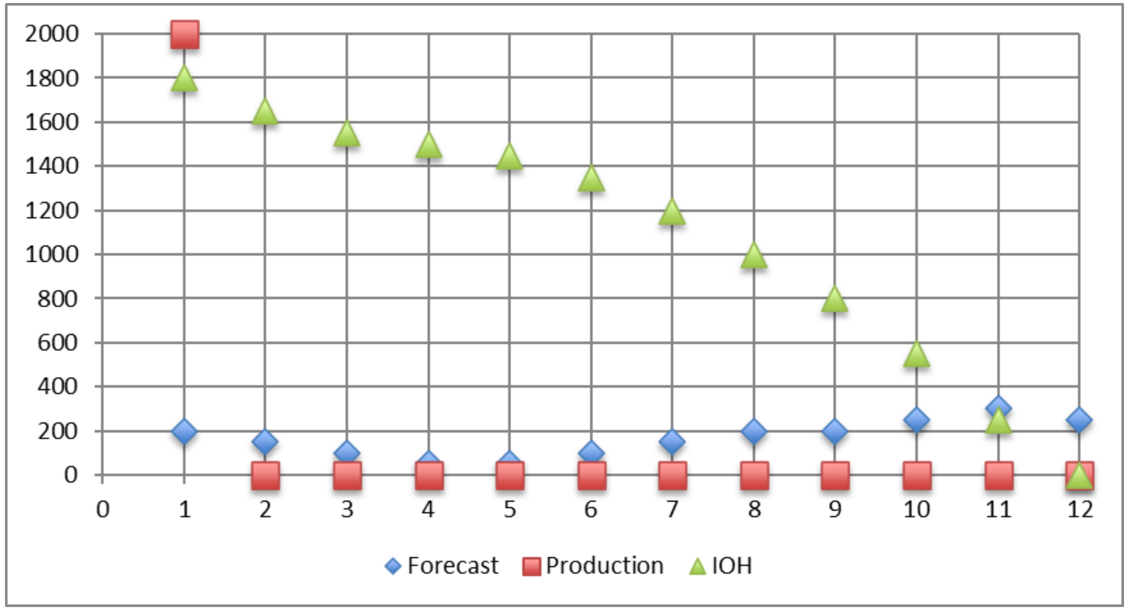
Optimal methods:
- Wagner-Within
- MILP

⇒Different ways to solve the problem
⇒We will look at the cost for each and determine the best

# Simple Heuristics: One Time Run

We will produce the 2000 modules the 1st month so that our production lines can be used for other products. It could be some form of ‘campaign’ loading.

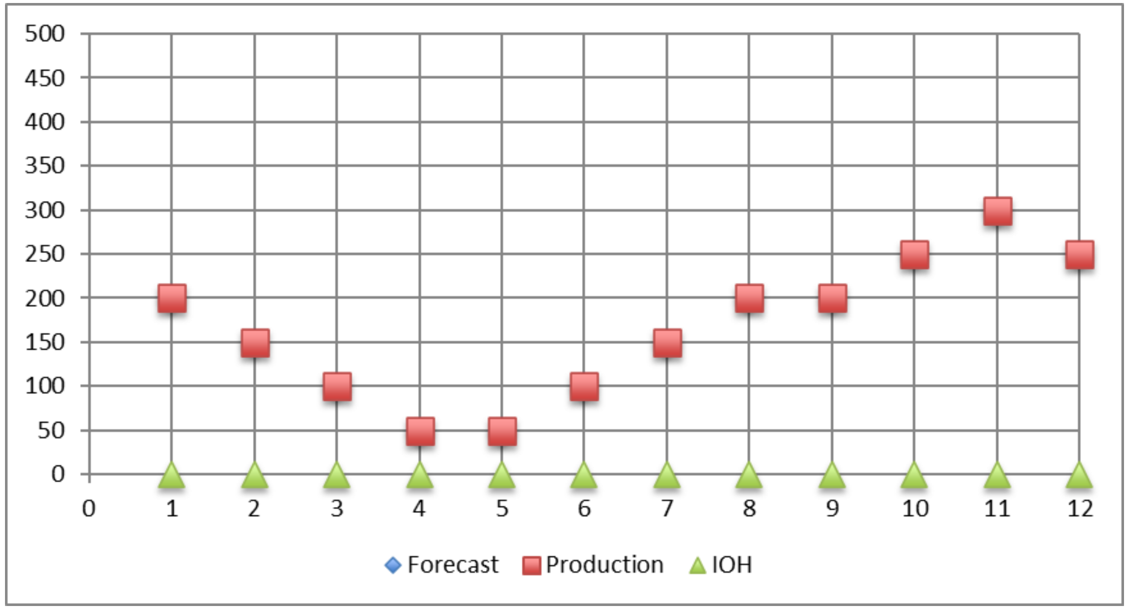
| Month | Forecast | Production | IOH | Holding Cost | Set-Up Costs | Total Cost |
|---|---|---|---|---|---|---|
| **1** | 200 | 2000 | 1800 | $ 1 800 | $ 500 | $ 2 300 |
| **2** | 150 | 0 | 1650 | $ 1 650 | $ - | $ 1 650 |
| **3** | 100 | 0 | 1550 | $ 1 550 | $ - | $ 1 550 |
| **4** | 50 | 0 | 1500 | $ 1 500 | $ - | $ 1 500 |
| **5** | 50 | 0 | 1450 | $ 1 450 | $ - | $ 1 450 |
| **6** | 100 | 0 | 1350 | $ 1 350 | $ - | $ 1 350 |
| **7** | 150 | 0 | 1200 | $ 1 200 | $ - | $ 1 200 |
| **8** | 200 | 0 | 1000 | $ 1 000 | $ - | $ 1 000 |
| **9** | 200 | 0 | 800 | $ 800 | $ - | $ 800 |
| **10** | 250 | 0 | 550 | $ 550 | $ - | $ 550 |
| **11** | 300 | 0 | 250 | $ 250 | $ - | $ 250 |
| 12 | 250 | 0 | 0 | $ - | $ - | $ - |
| **Total** | **2000** | **2000** | **13100** | **$ 13 100** | **$ 500** | **$ 13 600** |



# Simple Heuristics: Lot for Lot

We will what we need in the month we need it, following a chase strategy.
We incur no holding cost. We assume that our workers and equipment can be assigned to other products.

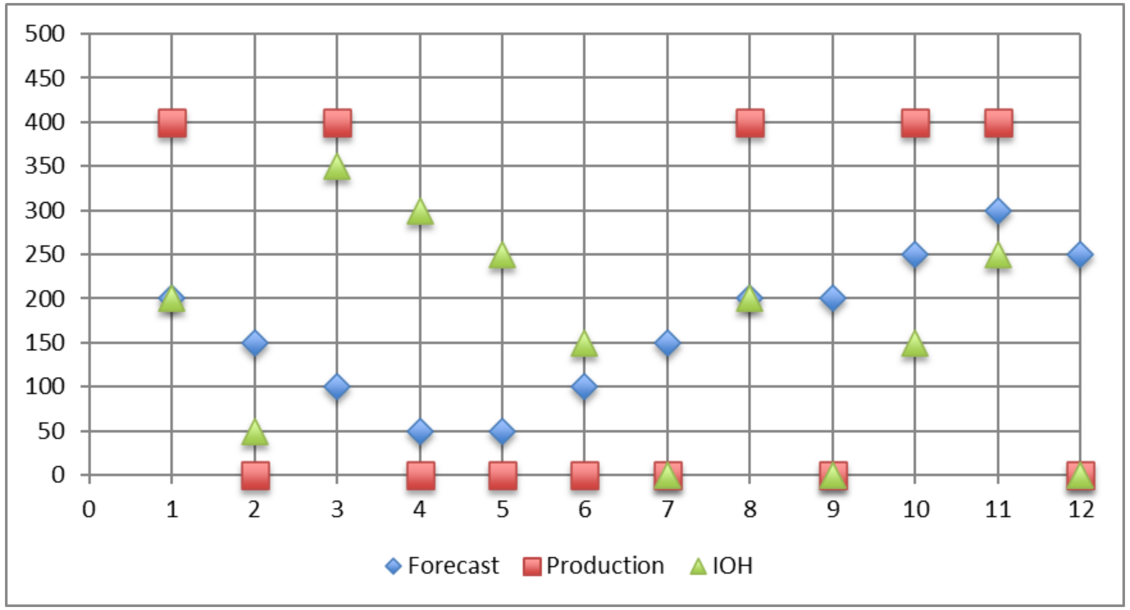
| Month | Forecast | Production | IOH | Holding Cost | Set-Up Costs | Total Cost |
|---|---|---|---|---|---|---|
| 1 | 200 | 200 | 0 | $ - | $ 500 | $ 500 |
| 2 | 150 | 150 | 0 | $ - | $ 500 | $ 500 |
| 3 | 100 | 100 | 0 | $ - | $ 500 | $ 500 |
| 4 | 50 | 50 | 0 | $ - | $ 500 | $ 500 |
| 5 | 50 | 50 | 0 | $ - | $ 500 | $ 500 |
| 6 | 100 | 100 | 0 | $ - | $ 500 | $ 500 |
| 7 | 150 | 150 | 0 | $ - | $ 500 | $ 500 |
| 8 | 200 | 200 | 0 | $ - | $ 500 | $ 500 |
| 9 | 200 | 200 | 0 | $ - | $ 500 | $ 500 |
| 10 | 250 | 250 | 0 | $ - | $ 500 | $ 500 |
| 11 | 300 | 300 | 0 | $ - | $ 500 | $ 500 |
| 12 | 250 | 250 | 0 | $ - | $ 500 | $ 500 |
| Total | 2000 | 2000 | 0 | $ - | $ 6 000 | $ 6 000 |



# Simple Heuristics: EOQ

We follow the demand but respect our EOQ to optimise the loading of our equipment. It might work well if you have multiple products to manufacture and run 'campaigns'.

| Month | Forecast | Production | IOH | Holding Cost | Set-Up Costs | Total Cost |
|---|---|---|---|---|---|---|
| 1 | 200 | 400 | 200 | $ 200 | $ 500 | $ 700 |
| 2 | 150 | 0 | 50 | $ 50 | $ - | $ 50 |
| 3 | 100 | 400 | 350 | $ 350 | $ 500 | $ 850 |
| 4 | 50 | 0 | 300 | $ 300 | $ - | $ 300 |
| 5 | 50 | 0 | 250 | $ 250 | $ - | $ 250 |
| 6 | 100 | 0 | 150 | $ 150 | $ - | $ 150 |
| 7 | 150 | 0 | 0 | $ - | $ - | $ - |
| 8 | 200 | 400 | 200 | $ 200 | $ 500 | $ 700 |
| 9 | 200 | 0 | 0 | $ - | $ - | $ - |
| 10 | 250 | 400 | 150 | $ 150 | $ 500 | $ 650 |
| 11 | 300 | 400 | 250 | $ 250 | $ 500 | $ 750 |
| 12 | 250 | 0 | 0 | $ - | $ - | $ - |
| Total | 2000 | 2000 | 1900 | $ 1 900 | $ 2 500 | $ 4 400 |



# Specialized Heuristics: Silver meal

Also called ‘least cost period’.
Very short-sighted approach. It does not consider the whole amount of information.
**Approach:** is it more cost effective to produce this month this month demand only or should I consider the following months as well? The decision is made to minimize the **average cost per period**, CT (Total Relevant Cost per period, T is the period over which span the orders).

Relevant cost is defined by :
- K, the set-up cost
- h:  holding cost per unit
- h*di+1 the holding cost for the demand of next period.

$C(T+j) = (K + h * d2$ +2*h*d3+… +(j-1)*h*dj)/(j+1)

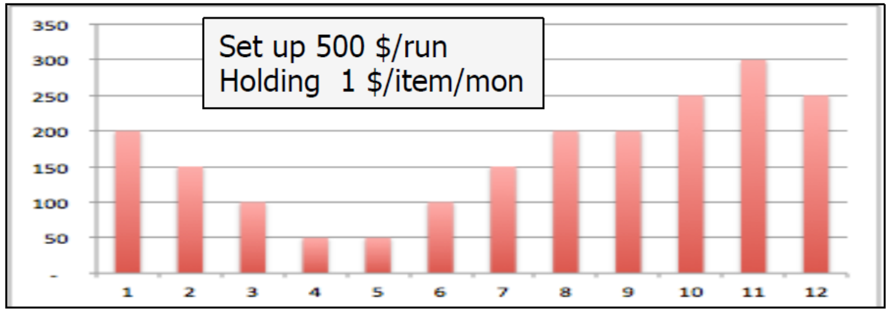
# Specialized Heuristics: Silver meal

$C(j) = (K + h * d2$ +2*h*d3+… +(j-1)*h*dj)/(j+1)

**Algorithm:**

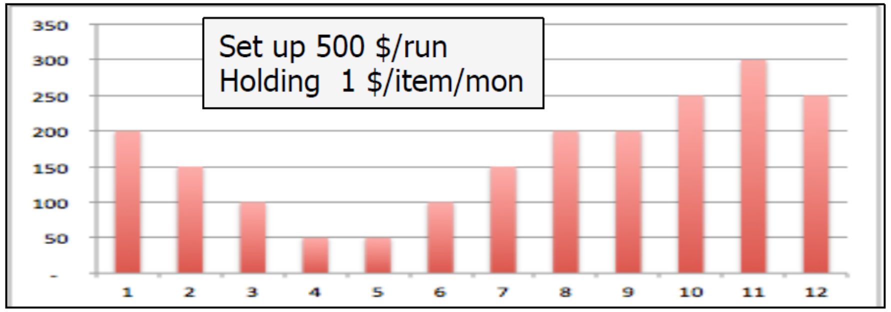
1. Start at T =1
2. Set j=0
3. Calculate C(T+j)
4. Set j= j+1
5. Calculate C(T+j)
6. If C(T+j) >C(T+j-1) :
   A. Place an order in time period T for the demand for period T to T+j-1
   B. Set R=T+j and go to step 2
7. Else go to step 4

# Specialized Heuristics: Silver meal




| | | | |
|---|---|---|---|
| T=1 | J=0 | C(1)= 500/1 | Make 200 in month 1 |
| T=1 | J=1 | C(2)= 500/2+150/2=325 | Make 350 in month 1, hold 150 for 1 month |
| T=1 | J=2 | C(3)= (500+150+2*100)/3=283 | Make 450 in month 1, hold 150 for 1 month, 100 for 2 months |
| T=1 | J=3 | C(4)= (500+150+2*100+3*50)/4=250 | Make 500 in month 1, hold 150 for 1 month, 100 for 2 months, 50 for 3m |
| T=1 | J=4 | C(5)= (500+150+2*100+3*50+4*50)/5=240 | Make 550 in month 1, hold 150 for 1 month, 100 for 2 months, 50 for 3m, 50 for 4m |
| T=1 | J=5 | C(6)= (500+150+2*100+3*50+4*50+5*100)/6=283 | C(6)>C(5) : we stop, make 550 in month 1 to cover demand for month 1 to 5; we restart in month 6 |

# Specialized Heuristics: Silver meal

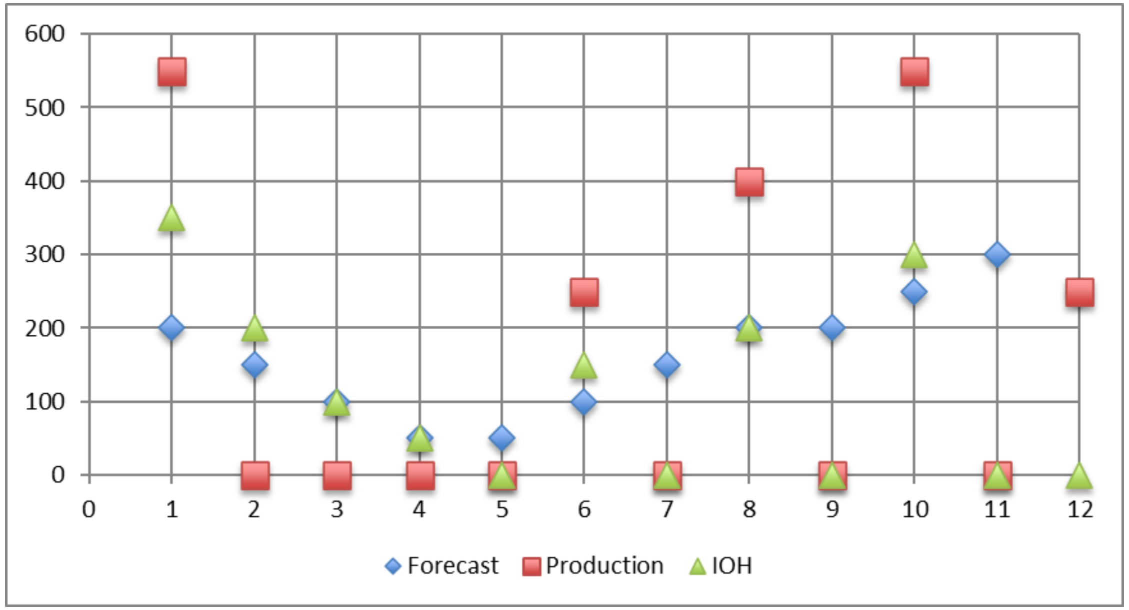



| | | | |
|---|---|---|---|
| T=6 | J=0 | C(6)= 500/1 | Make 100 |
| T=6 | J=1 | C(7)=(500+150)/2=325 | Make 250 in month 6, hold 150 for 1m |
| T=6 | J=2 | C(8)=(500+150+2*200)/3=350 | C(8)>C(7) : we stop, make 250 in month 6 to cover demand for month 6 and 7; we restart in month 8 |
| T=8 | J=0 | C(8)=500 | Make 200 |
| T=8 | J=1 | C(9)=(500+200)/2=350 | Make 400 in month 8, hold 200 for 1 month |
| T=8 | J=2 | C(10)=(500+200+250)/3=400 | C(10)>C(9) : we stop, make 400 in month 8 to cover demand for month 8and 9; we restart in month 10 |
| T=10 | J=0 | C(10)=500 | Make 250 |
| T=10 | J=1 | C(11)=(500+300)/2=400 | Make 550, hold 300 for 1 month |
| T=10 | J=2 | C(12)=(500+300+250*2)/3=433 | C(12)>C(11) : we stop, make 550 in month 10 to cover demand for month 10and 11; we restart in month 12 |

# Specialized Heuristics: Silver meal

Production Plan:
Month 1: 550
Month 6: 250
Month 8: 400
Month 10: 550
Month 12: 250

| Month | Forecast | Production | IOH | Holding Cost | Set-Up Costs | Total Cost |
|---|---|---|---|---|---|---|
| 1 | 200 | 550 | 350 | $ 350 | $ 500 | $ 850 |
| 2 | 150 | 0 | 200 | $ 200 | $ - | $ 200 |
| 3 | 100 | 0 | 100 | $ 100 | $ - | $ 100 |
| 4 | 50 | 0 | 50 | $ 50 | $ - | $ 50 |
| 5 | 50 | 0 | 0 | $ - | $ - | $ - |
| 6 | 100 | 250 | 150 | $ 150 | $ 500 | $ 650 |
| 7 | 150 | 0 | 0 | $ - | $ - | $ - |
| 8 | 200 | 400 | 200 | $ 200 | $ 500 | $ 700 |
| 9 | 200 | 0 | 0 | $ - | $ - | $ - |
| 10 | 250 | 550 | 300 | $ 300 | $ 500 | $ 800 |
| 11 | 300 | 0 | 0 | $ - | $ - | $ - |
| 12 | 250 | 250 | 0 | $ - | $ 500 | $ 500 |
| Total | 2000 | 2000 | 1350 | $ 1 350 | $ 2 500 | $ 3 850 |



# Optimizer: Wagner Within

It guarantees to find the minimum of set-up costs and inventory.

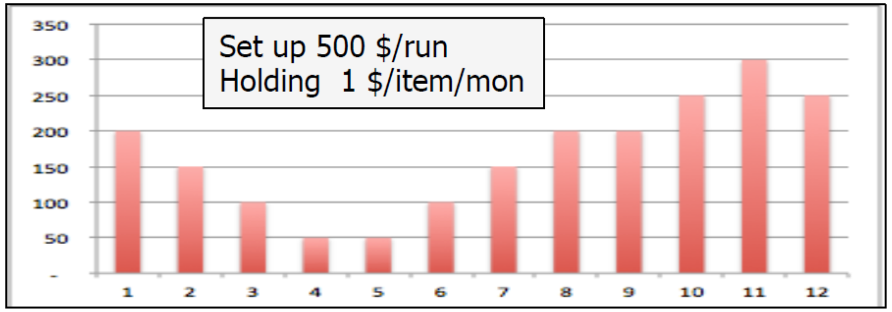
It relies on 2 properties:

- You only order when there is no inventory on hand, replenishments arrive at the beginning of the period, no shortage allowed and demand is deterministic
- There is an upper limit on holding time for demand when it exceeds set-up cost:

It is a forward backward algorithm:

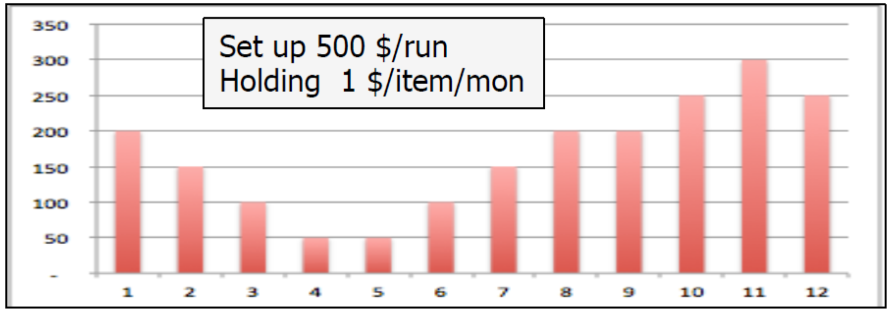
1. Start at time t
2. Find the cost for ordering just enough to satisfy demand in time t, D(t)
3. Look at all past orders up to t=1 and find cost for ordering D(t) as part of a past order for D(t-1)
4. Pick the lowest option
5. Proceed till the end of the period T
6. At t=T go backward finding the lowest cost option to produce D(t)

# Optimizer: Wagner Within




| | | |
|---|---|---|
| T=1 | Option 1: make 200 for D(1): TC(1)=500 | No other option |
| T=2 | Option 1: make 350 for D1 and D2: TC=650<br>Option 2: best plan for D1 and make D2: TC=1000 | Option 1 is best |
| T=3 | Option 1: make enough in m1 to cover D1, D2 and D3: TC= 850<br>Option 2: best T1 plan + make enough in T2 for D2 and D3: TC=500+500+100=1100<br>Option 3: best T2 plan + make enough for D3 TC=650+500=1150 | Option 1 is still the best |

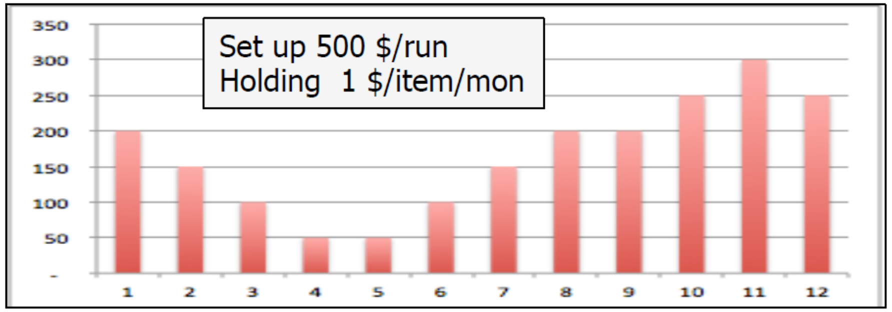
# Optimizer: Wagner Within




We continue with the same logic.

| | Period | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **Demand** | **200** | **150** | **100** | **50** | **50** | **100** | **150** | **200** | **200** | **250** | **300** | **250** |
| 1 | Order 1 | $500 | $650 | $850 | $1,000 | $1,200 | $1,700 | $2,600 | $4,000 | $5,600 | $7,850 | $10,850 | $13,600 |
| 2 | Order 2 | | $1,000 | $1,100 | $1,200 | $1,350 | $1,750 | $2,500 | $3,700 | $5,100 | $7,100 | $9,800 | $12,300 |
| 3 | Order 3 | | | $1,150 | $1,200 | $1,300 | $1,600 | $2,200 | $3,200 | $4,400 | $6,150 | $8,550 | $10,800 |
| 4 | Order 4 | | | | $1,350 | $1,400 | $1,600 | $2,050 | $2,850 | $3,850 | $5,350 | $7,450 | $9,450 |
| 5 | Order 5 | | | | | $1,500 | $1,600 | $1,900 | $2,500 | $3,300 | $4,550 | $6,350 | $8,100 |
| 6 | Order 6 | | | | | | $1,700 | $1,850 | $2,250 | $2,850 | $3,850 | $5,350 | $6,850 |
| 7 | Order 7 | | | | | | | $2,100 | $2,300 | $2,700 | $3,450 | $4,650 | $5,900 |
| 8 | Order 8 | | | | | | | | $2,350 | $2,550 | $3,050 | $3,950 | $4,950 |
| 9 | Order 9 | | | | | | | | | $2,750 | $3,000 | $3,600 | $4,350 |
| 10 | Order 10 | | | | | | | | | | $3,050 | $3,350 | $3,850 |
| 11 | Order 11 | | | | | | | | | | | $3,500 | $3,750 |
| 12 | Order 12 | | | | | | | | | | | | $3,850 |

# Optimizer: Wagner Within

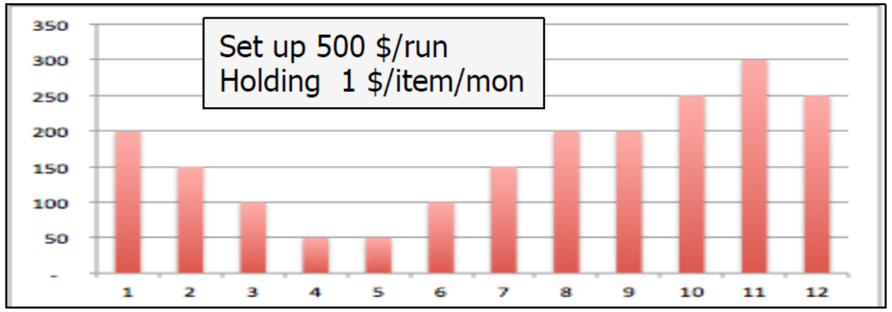



Backward analysis

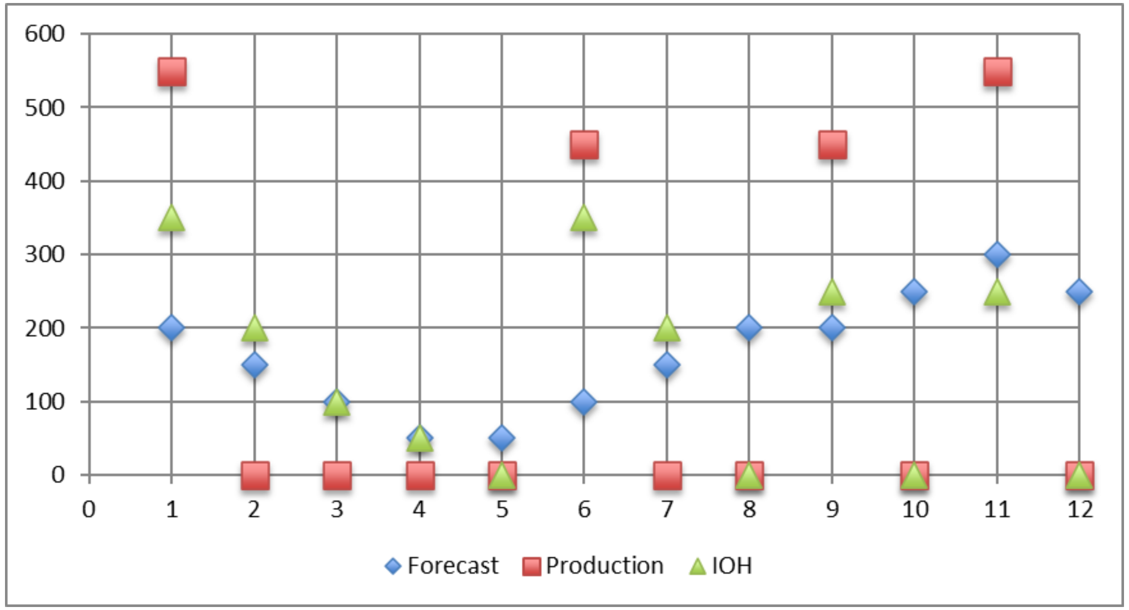
| | Period | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Demand | 200 | 150 | 100 | 50 | 50 | 100 | 150 | 200 | 200 | 250 | 300 | 250 |
| 1 | Order 1 | $500 | $650 | $850 | $1,000 | $1,200 | $1,700 | $2,600 | $4,000 | $5,600 | $7,850 | $10,850 | $13,600 |
| 2 | Order 2 | | $1,000 | $1,100 | $1,200 | $1,350 | $1,750 | $2,500 | $3,700 | $5,100 | $7,100 | $9,800 | $12,300 |
| 3 | Order 3 | | | $1,150 | $1,200 | $1,300 | $1,600 | $2,200 | $3,200 | $4,400 | $6,150 | $8,550 | $10,800 |
| 4 | Order 4 | | | | $1,350 | $1,400 | $1,600 | $2,050 | $2,850 | $3,850 | $5,350 | $7,450 | $9,450 |
| 5 | Order 5 | | | | | $1,500 | $1,600 | $1,900 | $2,500 | $3,300 | $4,550 | $6,350 | $8,100 |
| 6 | Order 6 | | | | | | $1,700 | $1,850 | $2,250 | $2,850 | $3,850 | $5,350 | $6,850 |
| 7 | Order 7 | | | | | | | $2,100 | $2,300 | $2,700 | $3,450 | $4,650 | $5,900 |
| 8 | Order 8 | | | | | | | | $2,350 | $2,550 | $3,050 | $3,950 | $4,950 |
| 9 | Order 9 | | | | | | | | | $2,750 | $3,000 | $3,600 | $4,350 |
| 10 | Order 10 | | | | | | | | | | $3,050 | $3,350 | $3,850 |
| 11 | Order 11 | | | | | | | | | | | $3,500 | $3,750 |
| 12 | Order 12 | | | | | | | | | | | | $3,850 |

**Optimal Production Plan:**
**Make 550 in period 1, 450 in period 6, 450 in period 9, and 550 in period 11**

# Optimizer: Wagner Within




| Month | Forecast | Production | IOH | Holding Cost | Set-Up Costs | Total Cost |
|---|---|---|---|---|---|---|
| 1 | 200 | 550 | 350 | $ 350 | $ 500 | $ 850 |
| 2 | 150 | 0 | 200 | $ 200 | $ - | $ 200 |
| 3 | 100 | 0 | 100 | $ 100 | $ - | $ 100 |
| 4 | 50 | 0 | 50 | $ 50 | $ - | $ 50 |
| 5 | 50 | 0 | 0 | $ - | $ - | $ - |
| 6 | 100 | 450 | 350 | $ 350 | $ 500 | $ 850 |
| 7 | 150 | 0 | 200 | $ 200 | $ - | $ 200 |
| 8 | 200 | 0 | 0 | $ - | $ - | $ - |
| 9 | 200 | 450 | 250 | $ 250 | $ 500 | $ 750 |
| 10 | 250 | 0 | 0 | $ - | $ - | $ - |
| 11 | 300 | 550 | 250 | $ 250 | $ 500 | $ 750 |
| 12 | 250 | 0 | 0 | $ - | $ - | $ - |
| Total | 2000 | 2000 | 1750 | $ 1 750 | $ 2 000 | $ 3 750 |



# Optimizer: MILP

### Decision Variables

$$Q_t = \text{Quantity produced in time period t (units)} \quad \forall t$$

$$Z_t = 1 \text{ if manufacture in time period t, } = 0 \text{ o.w.} \quad \forall t$$

### Input Data

$$CAP_t = \text{Manufacturing capacity for time period t (units)} \quad \forall t \in T$$

$$D_t = \text{Demand for time period t (units)} \quad \forall t \in T$$

$$I_t = \text{Inventory at end of time period t (units)} \quad \forall t \in T$$

$$c_{setup} = \text{Cost to make a production run (\$/run)}$$

$$h = \text{Inventory holding cost for one time period (\$/unit/time period)}$$

$$M = \text{Large number}$$

# Optimizer: MILP

Set up Costs

$$Min \quad z = \boxed{\sum_t c_{setup} Z_t} + \boxed{\sum_t h I_t}$$

Holding Costs

$s.t.$

$$I_0 = 0$$

$$\boxed{Q_t - D_t + I_{t-1} - I_t = 0 \quad \forall t \in T}$$

Conservation of Flow or Inventory Balance constraint for time period t.

$$\boxed{MZ_t - Q_t \geq 0 \quad \forall t \in T}$$

Linking/logical constraint ensuring binary flag for a time period is ON if we manufacture any quantity during that time period.

$$\boxed{Q_t \leq CAP_t \quad \forall t \in T}$$

Maximum capacity constraint for time period t.

$$I_t, Q_t \geq 0 \quad \forall t \in T$$

$$Z_t = \{0,1\} \quad \forall t \in T$$

# Optimizer: MILP

FPH MILP

| | |
|---|---|
| | Cells can be changed |
| | Decision Variables |
| | DO NOT CHANGE - formula |

| | | |
|---|---|---|
| Lead Time | 0 | |
| Setup | $ 500 | $/run |
| Holding | $ 1.00 | $/item/month |
| M | 2,000 | |
| Capacity | 99999 | units/month |

| | |
|---|---|
| $ 2,000 | Set Up Cost |
| $ 1,750 | Holding Cost |
| $ 3,750 | Total |

| Period | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Demand | 200 | 150 | 100 | 50 | 50 | 100 | 150 | 200 | 200 | 250 | 300 | 250 |
| Beg Inv | 0 | 350 | 200 | 100 | 50 | 0 | 350 | 200 | 0 | 250 | 0 | 250 |
| Quantity to Make | 550 | 0 | 0 | 0 | 0 | 450 | 0 | 0 | 450 | 0 | 550 | 0 |
| Order Setup | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 |
| Capacity | 99999 | 99999 | 99999 | 99999 | 99999 | 99999 | 99999 | 99999 | 99999 | 99999 | 99999 | 99999 |
| End Inv | 350 | 200 | 100 | 50 | 0 | 350 | 200 | 0 | 250 | 0 | 250 | 0 |

| Period | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Setup cost | $ 500 | $ - | $ - | $ - | $ - | $ 500 | $ - | $ - | $ 500 | $ - | $ 500 | $ - |
| Holding cost | $ 350 | $ 200 | $ 100 | $ 50 | $ 0 | $ 350 | $ 200 | $ - | $ 250 | $ - | $ 250 | $ - |
| Totalcost | $ 850 | $ 200 | $ 100 | $ 50 | $ 0 | $ 850 | $ 200 | $ - | $ 750 | $ - | $ 750 | $ - |

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| logical constraints | 1450 | 0 | 0 | 0 | 0 | 1550 | 0 | 0 | 1550 | 0 | 1450 | 0 |

# Comparison

The most complex algorithm does not automatically yield the best results.
Simple heuristics are easy to implement
Specialty heuristics are more advanced but harder to setup.
Optimised methods require more time and data. They need a good forecast and treat equally today vs in 6 months dat
Silvermeal is good when the forecast shows large swings from month to month.

| **Month** | **Demand** | Single Run | Lot for Lot | EOQ | SM | WW | MILP |
|---|---|---|---|---|---|---|---|
| 1 | 200 | 2000 | 200 | 400 | 550 | 550 | 550 |
| 2 | 150 | 0 | 150 | 0 | 0 | 0 | 0 |
| 3 | 100 | 0 | 100 | 400 | 0 | 0 | 0 |
| 4 | 50 | 0 | 50 | 0 | 0 | 0 | 0 |
| 5 | 50 | 0 | 50 | 0 | 0 | 0 | 0 |
| 6 | 100 | 0 | 100 | 0 | 250 | 450 | 450 |
| 7 | 150 | 0 | 150 | 0 | 0 | 0 | 0 |
| 8 | 200 | 0 | 200 | 400 | 400 | 0 | 0 |
| 9 | 200 | 0 | 200 | 0 | 0 | 450 | 450 |
| 10 | 250 | 0 | 250 | 400 | 550 | 0 | 0 |
| 11 | 300 | 0 | 300 | 400 | 0 | 550 | 550 |
| 12 | 250 | 0 | 250 | 0 | 250 | 0 | 0 |
| Total: | 2 000 | 2000 | 2 000 | 2000 | 2000 | 2000 | 2000 |
| | | | | | | | |
| Holding Cost | | 13100 | | 1900 | 1350 | 1750 | 1750 |
| Order cost | | 500 | 6 000 | 2500 | 2500 | 2000 | 2000 |
| **Total cost** | | **13600** | **6 000** | **4400** | **3850** | **3750** | **3750** |
| Deviation to Opt | | 263% | 60% | 17% | 3% | | |

# Wrap-up

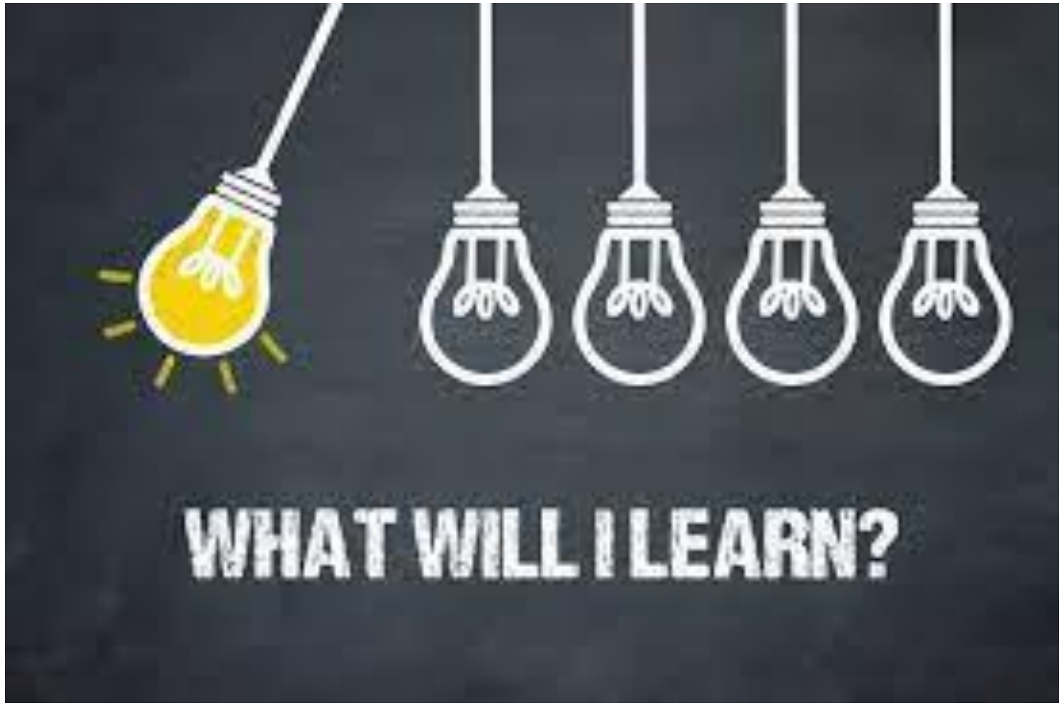
- The MPS runs at SKU level and focus on the next 3 to 6 months
- The levers during the MPS horizon are limited
- The MPS is an input for the next process to trigger production order and material purchase
- Different methods can be considered to develop the master production schedule
- Coherence with the aggregate plan needs to be validated

# Agenda Course 4

- MPS
  - Introduction
  - Simple Heuristics: lot for lot/fixed EOQ
  - Specialized heuristics

- MRP: Material Requirement Planning
  - BOM: Bill of Material
  - MRP
  - Sequential vs simultaneous optimization

# Learning Objectives




**Understand** dependent demand vs independent demand

**Develop** a product nomenclature

**Build** a gross and net requirement plan

**Evaluate** the influence of lot size

**Describe** an ERP

# MPS to MRP

**Objective**: Compute from the MPS the needs in terms of components and raw materials (Material Requirements Planning)

Provide Production/Manufacturing Orders (MO) staggered over time (dates and quantities) allowing to satisfy Customer Needs (or independent demands)

Provide Purchase Orders (PO) staggered over time (dates and quantities) allowing orders to be placed with suppliers

# Purpose of MRP

- What to order
- When to order
- How much to order
- When is the item needed

# Dependent demand

The demand for one item is related to the demand for another item

Given a quantity for the end item, the demand for all parts and components can be calculated

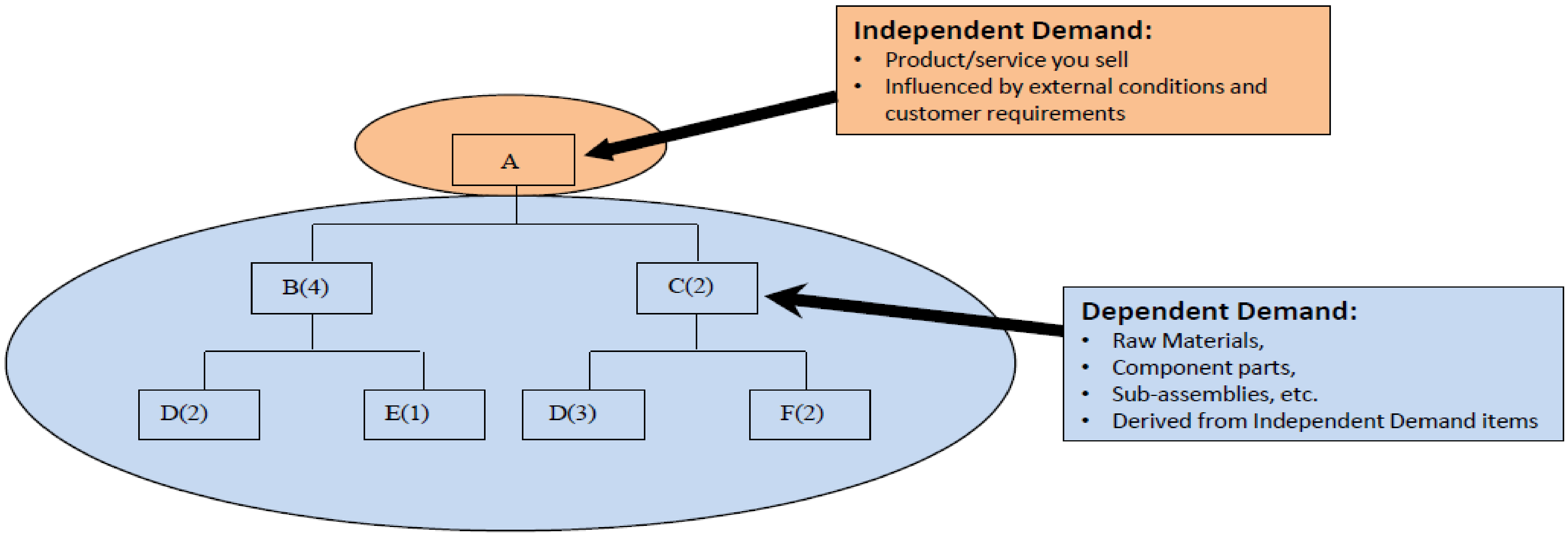
In general, used whenever a schedule can be established for an item

MRP : a dependent demand technique that uses a bill-of-material, inventory, expected receipts, lead time and a master of production schedule to determine material requirements.

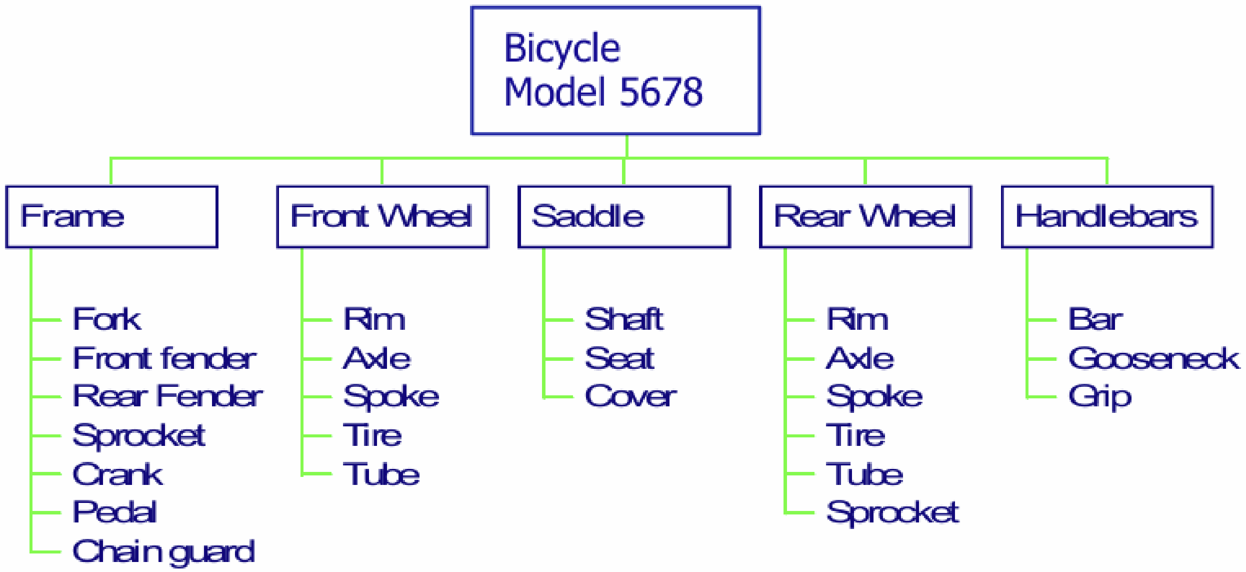
# BOM

**The bill of materials (BoM) is a list of the parts or components that are required to build a product**

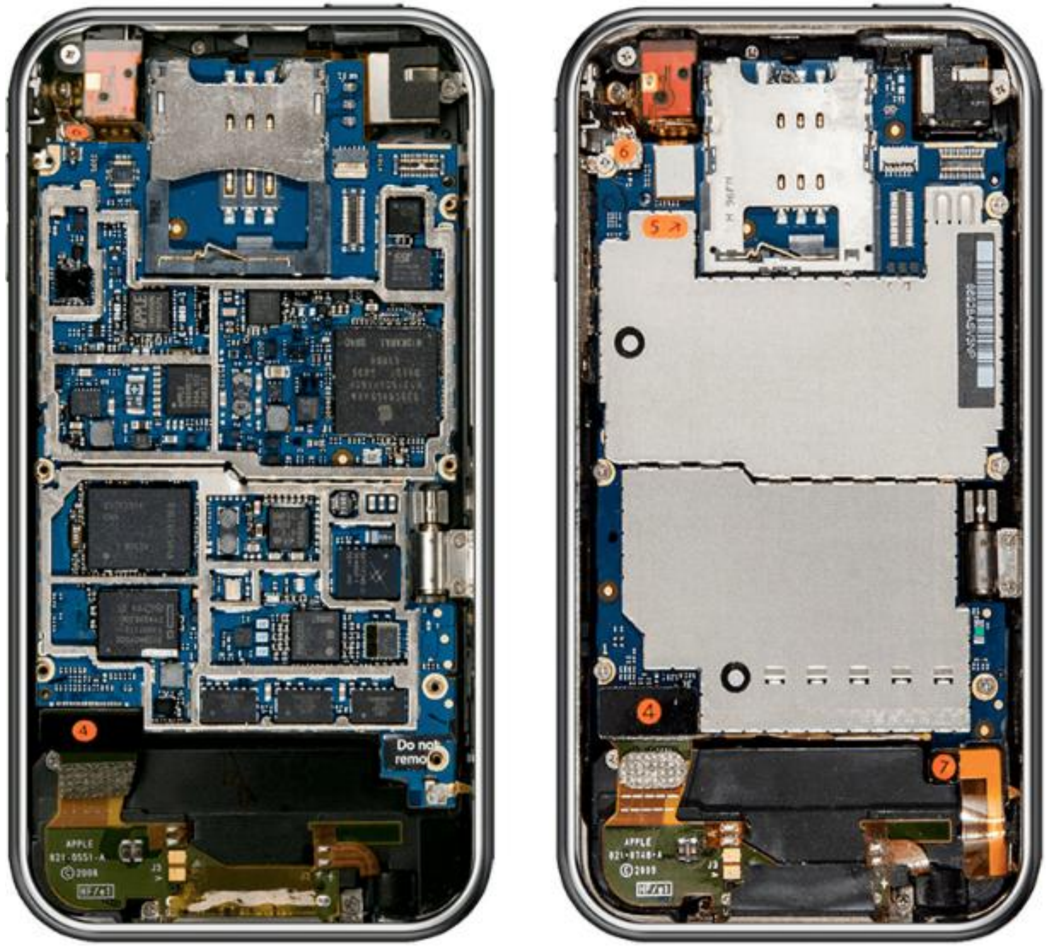
- WhatIs.com Definition



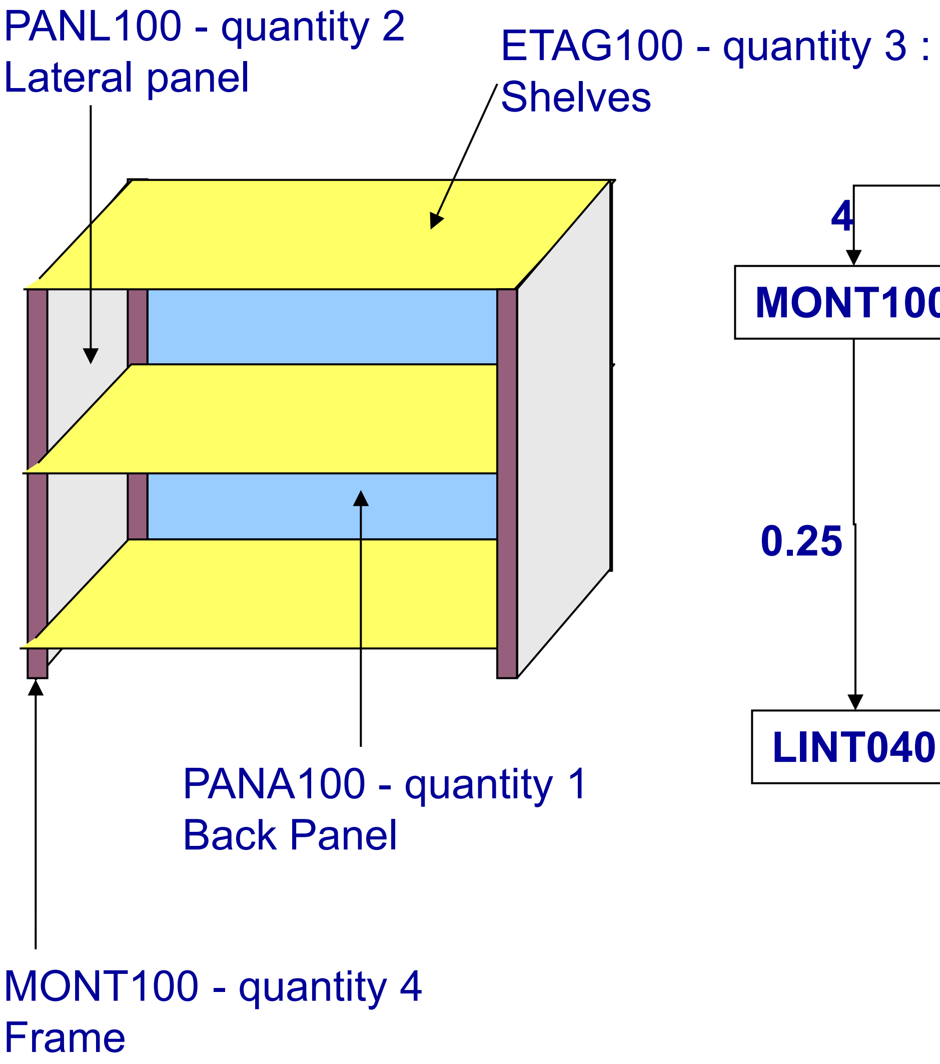
# Multi-tiered BOM

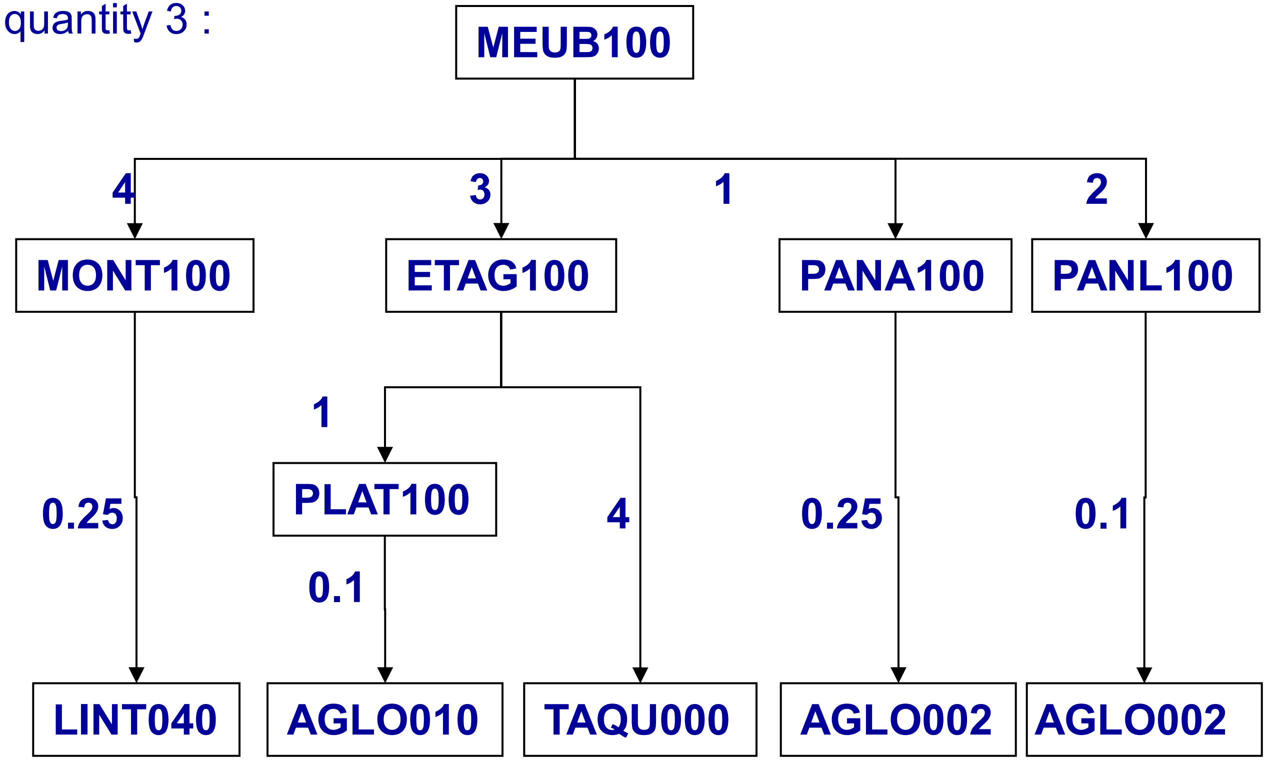



# Complex BOM

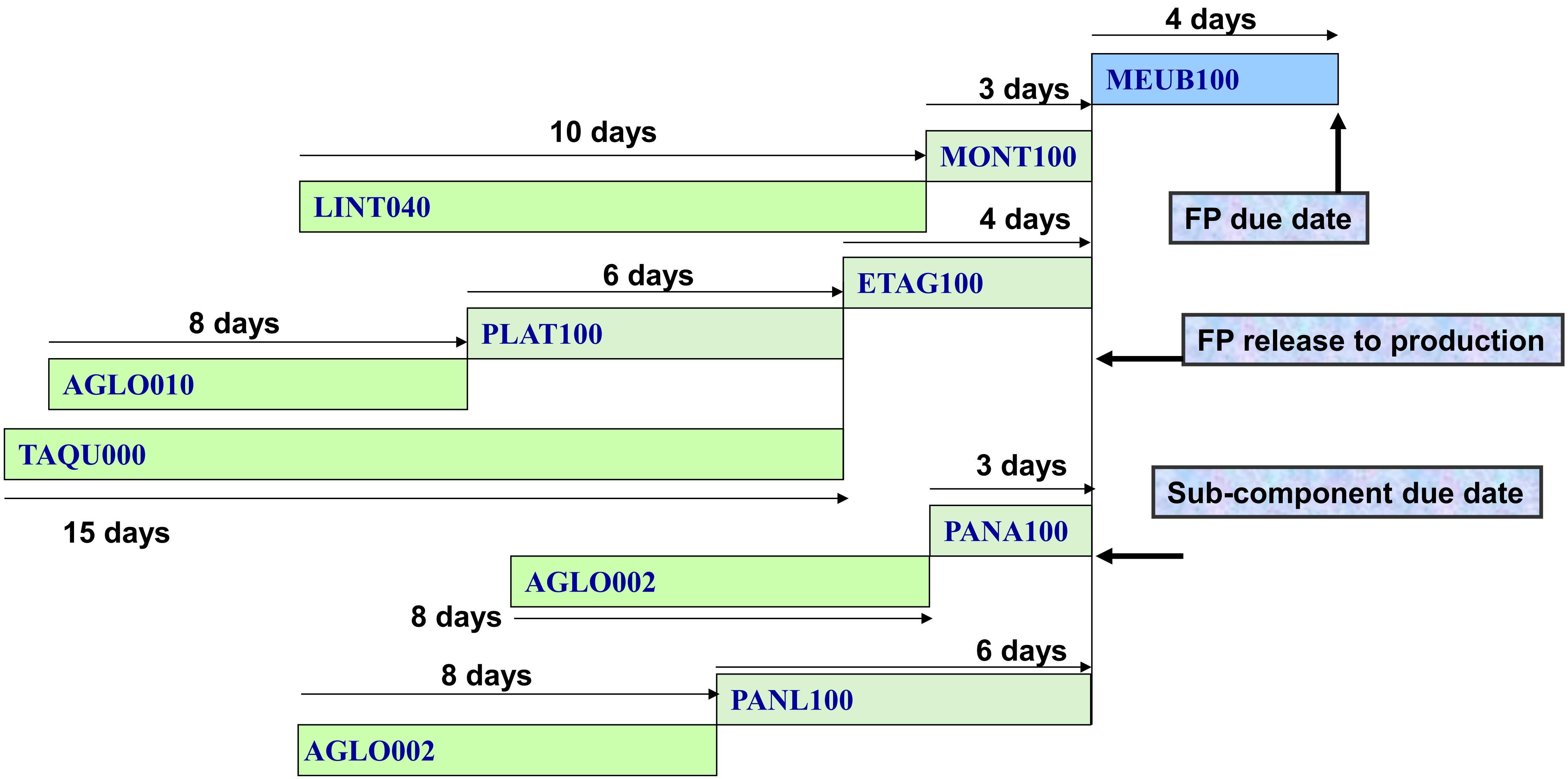


# BOM example


PANL100 - quantity 2
Lateral panel
ETAG100 - quantity 3 :
Shelves
PANA100 - quantity 1
Back Panel
MONT100 - quantity 4
Frame


MEUB100
4
3
1
2
MONT100
ETAG100
PANA100
PANL100
1
PLAT100
0.25
4
0.25
0.1
0.1
LINT040
AGLO010
TAQU000
AGLO002
AGLO002

# Time based explosion



# MRP assumption

Demand is known
No back-ordering
No capacity constraints
Static planning over horizon
Lead-times known, constant and independent of lot-size

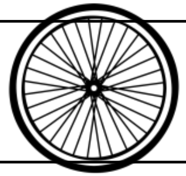
# Example 2

FP due date

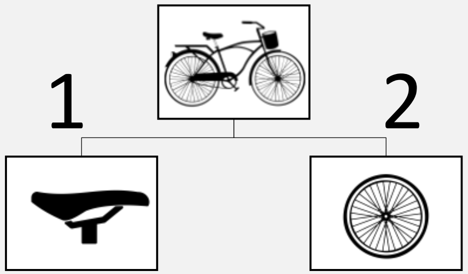
| Période | … | 4 | 5 | 6 | 7 |
| --- | --- | --- | --- | --- | --- |
|  | … | 5 | 5 | 7 | 10 |

Production LT : 2 weeks

FP start date

| Période | … | 4 | 5 | 6 | 7 |
| --- | --- | --- | --- | --- | --- |
|  | | | | 7 | |
|  | | 14 | | | |
|  | | 7 | | | |

How many wheels and saddles do we need to deliver 7 bikes in week 6



Nomenclature

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# Example 2

FP due date

| Période | … | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|
|  | … | 5 | 5 | 7 | 10 |

Production LT : 2 weeks

FP start date

| Période | … | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|
|  | … | | | 7 | 10 |
| | | 14 | | | |
| | | 7 | | | |

**Net requirements**

| | |
|---|---|
| Gross needs | 14 |
| In stock | (6) |
| On order | (4) |
| **Net demand** | **4** |

**Stock** 

**6** : in store in week 4

New order of 4 with due date in week 4

**4** : to receive in week 4

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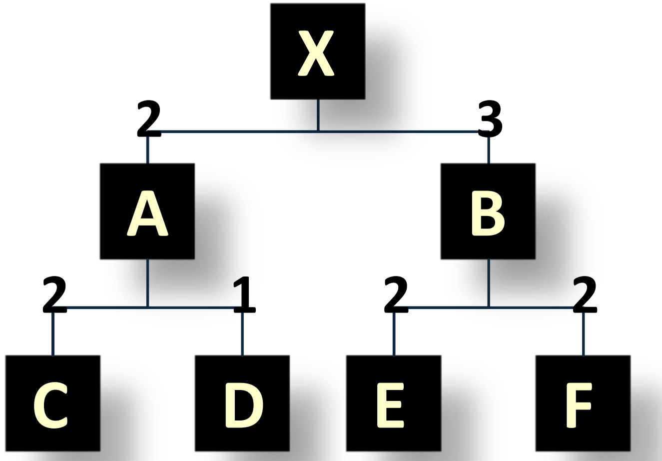
# Explosion calculus

Rules for translating gross requirements at one level to production schedule at that level and requirements at lower levels.

Basic Algorithm

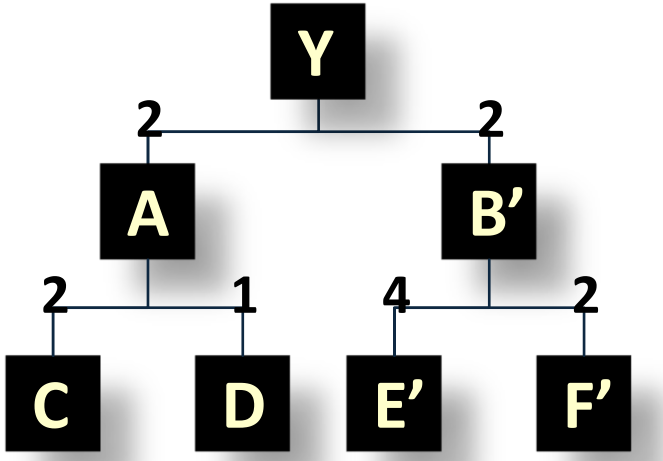
1. Start with level i: the demand (i=0)
2. Find Gross requirements (GR) and On Hand inventory for level i
3. Find Net Requirements (NR) for level i: Net Req. = GR- Scheduled Receipts-OH-inventory
4. Establish Planned Order Release (POR) for Level i using the relevant Lead Times
5. Set GR for Level i+1 based on POR for Level i
6. Set i=i+1 and go to step 2

# Example




| | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| X | 5 | 3 | 6 | 8 | 6 |
| Y | 2 | 5 | 5 | 4 | 3 |

FP Gross requirements based on MPS




| | Stock | LT |
|---|---|---|
| X | 7 | 1 |
| Y | 4 | 1 |

| | Stock | LT |
|---|---|---|
| A | 30 | 1 |
| B | 25 | 1 |
| B’ | 18 | 1 |

| | Stock | LT |
|---|---|---|
| C | 90 | 1 |
| D | 50 | 1 |
| E | 80 | 1 |
| F | 80 | 1 |
| E’ | 85 | 1 |
| F’ | 23 | 1 |

# Example lot for lot

| X | | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Gross requirements | | 5 | 3 | 6 | 8 | 6 |
| On hand inv | 7 | | | | | |
| Net requirements | | | | | | |
| Production order | | | | | | |

| Y | | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Gross requirements | | 2 | 5 | 5 | 4 | 3 |
| On hand inv | 4 | | | | | |
| Net requirements | | | | | | |
| Production order | | | | | | |

# Example

| X | | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Gross demand | | 5 | 3 | 6 | 8 | 6 |
| On hand inv | 7 | 2 | 0 | 0 | 0 | 0 |
| Net demand | | | 1 | 6 | 8 | 6 |
| On order | | 1 | 6 | 8 | 6 | |

| B | | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Gross requirements | | 3 | 18 | 24 | 18 | |
| Inventory | 25 | 22 | 4 | 0 | | |
| Net requirements | | 0 | 0 | 20 | 18 | |
| Orders | | | 20 | 18 | | |

# Your Turn

| X | | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Gross requirements | | 5 | 3 | 6 | 8 | 6 |
| On hand inv | 7 | | | | | |
| Net requirements | | | | | | |
| Production order | | | | | | |

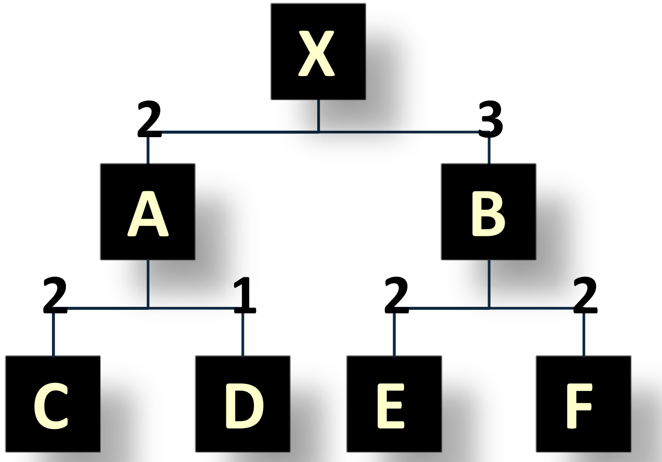
| Y | | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Gross requirements | | 2 | 5 | 5 | 4 | 3 |
| On hand inv | 4 | | | | | |
| Net requirements | | | | | | |
| Production order | | | | | | |

# Your Turn

| A | | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Gross requirements | | | | | | |
| On hand inv | 30 | | | | | |
| Net requirements | | | | | | |
| Purchase order | | | | | | |

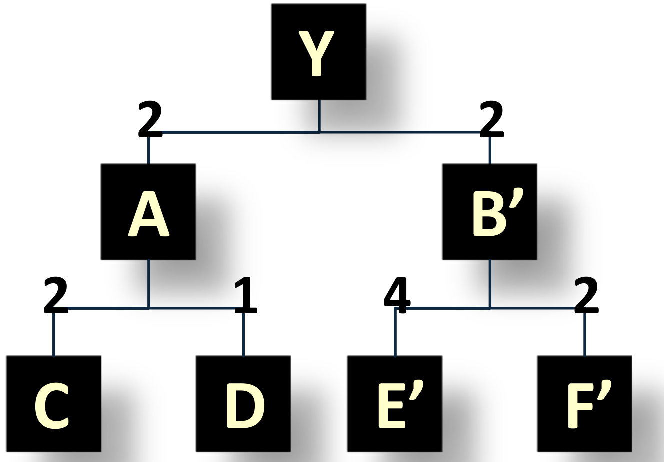
| B | | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Gross requirements | | | | | | |
| On hand inv | 25 | | | | | |
| Net requirements | | | | | | |
| Purchase order | | | | | | |

# Your Turn EOQ




| | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| X | 5 | 3 | 6 | 8 | 6 |
| Y | 2 | 5 | 5 | 4 | 3 |

Gross requirements




| | Stock | LT | EOQ |
|---|---|---|---|
| X | 7 | 1 | 5 |
| Y | 4 | 1 | 5 |

| | Stock | LT | EOQ |
|---|---|---|---|
| A | 30 | 1 | 15 |
| B | 25 | 1 | 15 |
| B’ | 18 | 1 | 15 |

| | Stock | LT | EOQ |
|---|---|---|---|
| C | 90 | 1 | 30 |
| D | 50 | 1 | 30 |
| E | 80 | 1 | 30 |
| F | 80 | 1 | 30 |
| E’ | 85 | 1 | 30 |
| F’ | 23 | 1 | 30 |

# Your Turn EOQ

| X | | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Gross requirements | | 5 | 3 | 6 | 8 | 6 |
| On hand inv | 7 | | | | | |
| Net requirements | | | | | | |
| Production order | | | | | | |

| Y | | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Gross requirements | | 2 | 5 | 5 | 4 | 3 |
| On hand inv | 4 | | | | | |
| Net requirements | | | | | | |
| Production order | | | | | | |

# Your Turn

| A | | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Gross requirements | | | | | | |
| On hand inv | 30 | | | | | |
| Net requirements | | | | | | |
| Purchase order | | | | | | |

| B | | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Gross requirements | | | | | | |
| On hand inv | 25 | | | | | |
| Net requirements | | | | | | |
| Purchase order | | | | | | |

# Your Turn

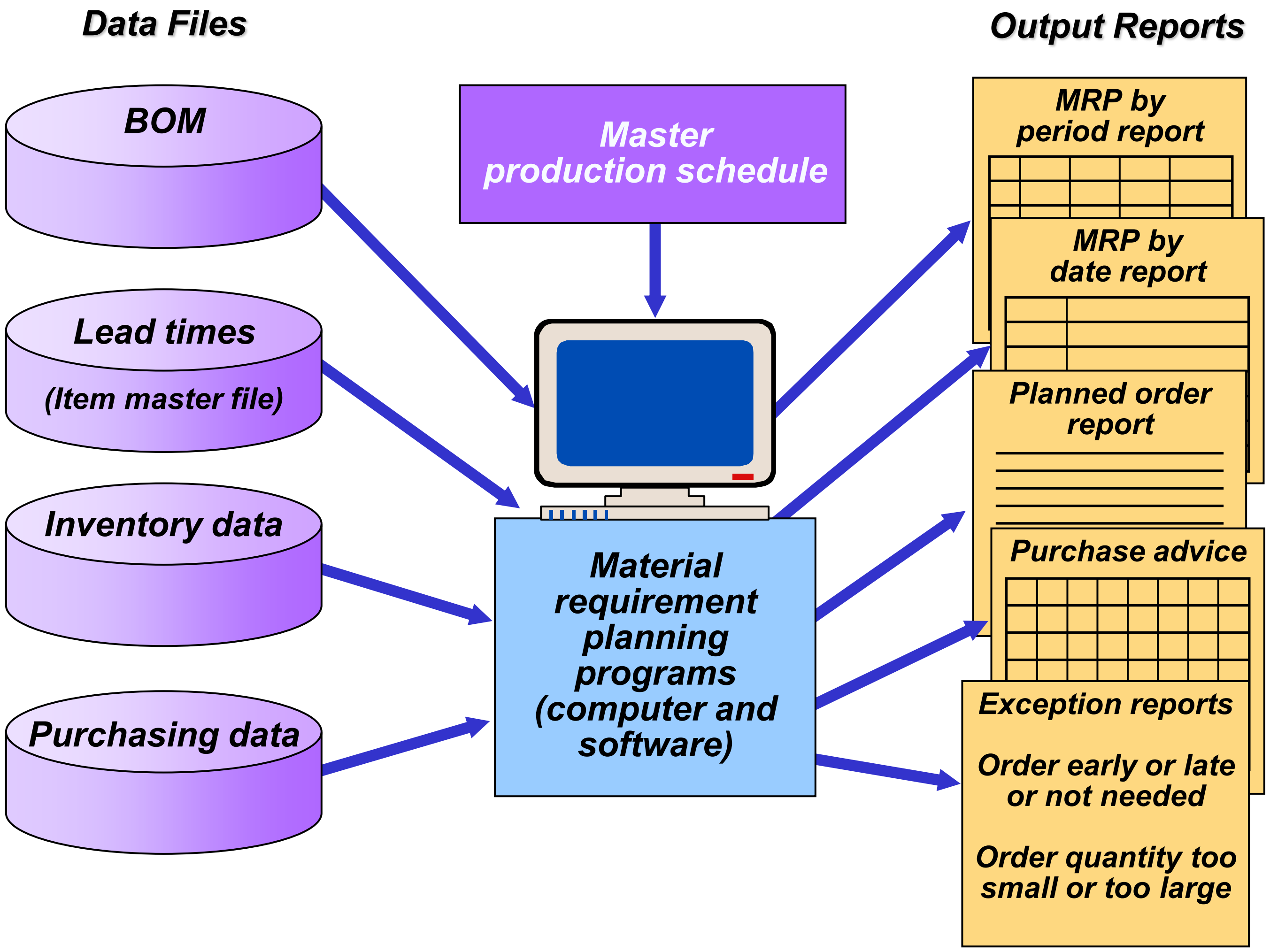
| A | | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Gross requirements | | 20 | 20 | 20 | 20 | |
| On hand inv | 30 | 10 | 5 | 0 | 10 | |
| Net requirements | | | 10 | 15 | 20 | |
| Purchase order | | 15 | 15 | 30 | | |

| B | | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Gross requirements | | 15 | 15 | 15 | 30 | |
| On hand inv | 25 | 10 | 10 | 10 | 10 | |
| Net requirements | | | 5 | 5 | 20 | |
| Purchase order | | 15 | 15 | 30 | | |

# MRP Limitations

Supplier might be delayed or LT might not be accurate.
Transportation might be delayed.
If one component is delayed the schedule of all others is pushed out.
Production might have some short fall due to quality issue
=> Production yield needs to be factored in.

# MRP structure



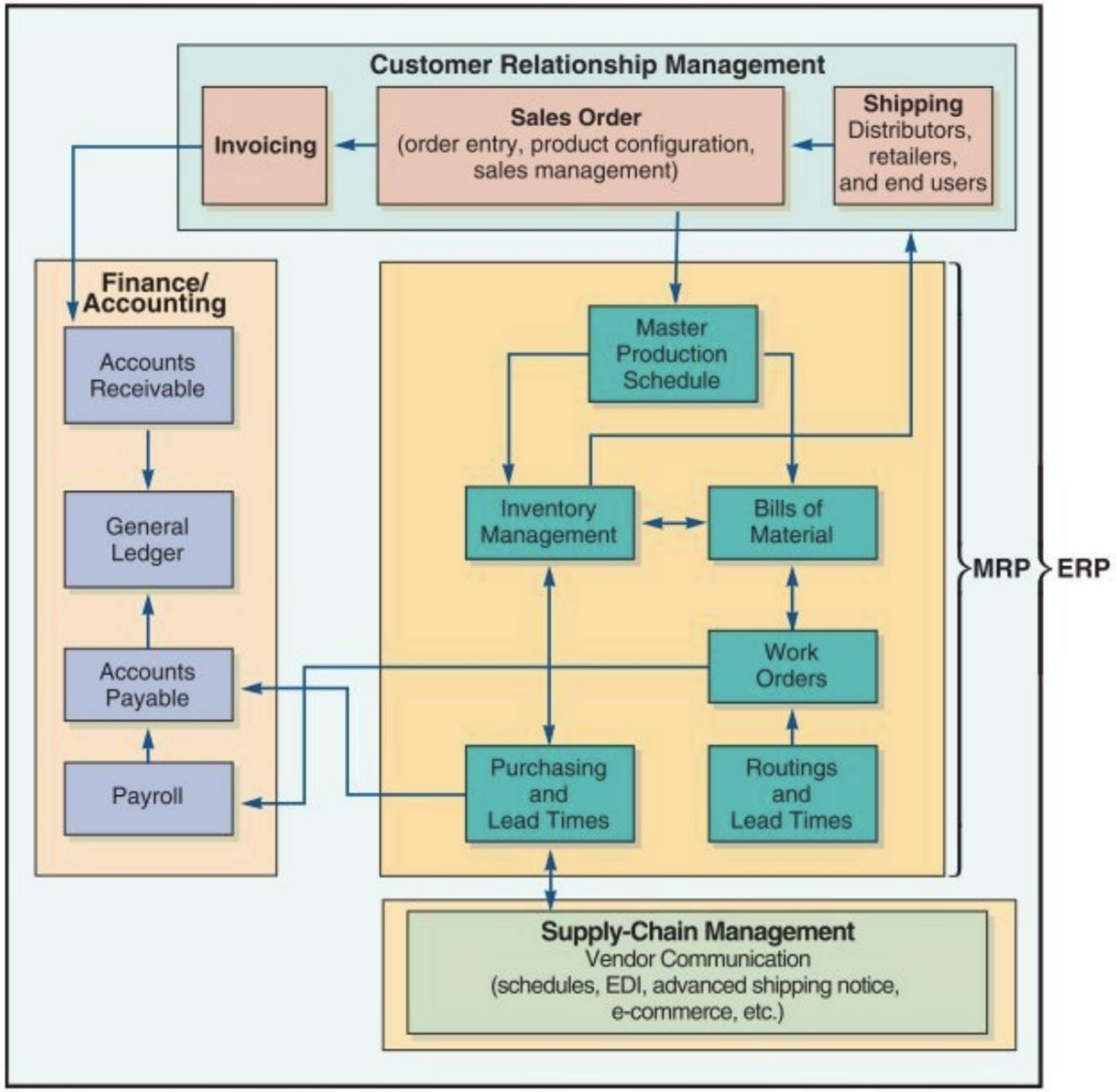
# Material Requirements Planning II

- ☑ Once an MRP system is in place, inventory data can be augmented by other useful information
  - ☑ Labor hours
  - ☑ Material costs
  - ☑ Capital costs
  - ☑ Virtually any resource
- ☑ System is generally called MRP II or Material Resource Planning: incorporated marketing, finance, accounting, engineering, and human resources aspects into the planning process.

# ERP

- ☑ An extension of the MRP system to tie in customers and suppliers
    1. Allows automation and integration of many business processes
    2. Shares common data bases and business practices
    3. Produces information in real time
- ☑ Coordinates business from supplier evaluation to customer invoicing
- ☑ ERP modules include
    - ☑ Basic MRP
    - ☑ Finance
    - ☑ Human resources
    - ☑ Supply chain management (SCM)
    - ☑ Customer relationship management (CRM)
- ☑ ERP systems have the potential to
    - ☑ Reduce transaction costs
    - ☑ Increase the speed and accuracy of information

# ERP modules

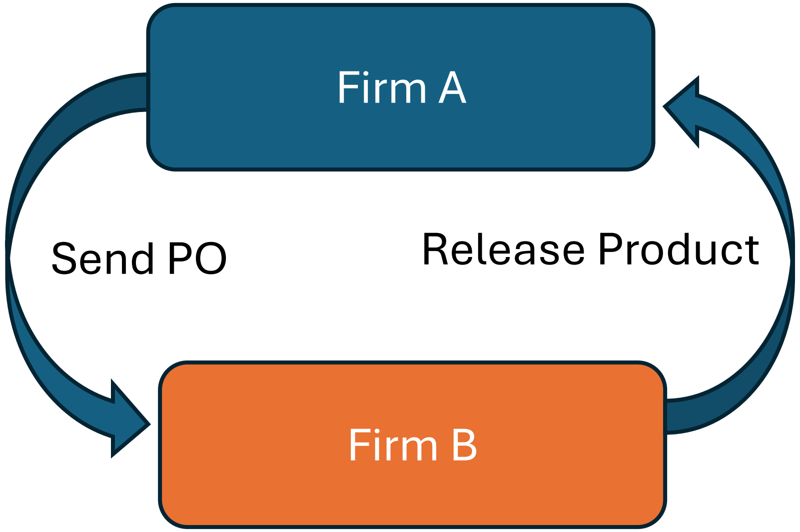

Customer Relationship Management
Invoicing
Sales Order
(order entry, product configuration, sales management)
Shipping
Distributors, retailers, and end users
Finance/ Accounting
Accounts Receivable
General Ledger
Accounts Payable
Payroll
Master Production Schedule
Inventory Management
Bills of Material
Work Orders
Purchasing and Lead Times
Routings and Lead Times
MRP
ERP
Supply-Chain Management
Vendor Communication
(schedules, EDI, advanced shipping notice, e-commerce, etc.)

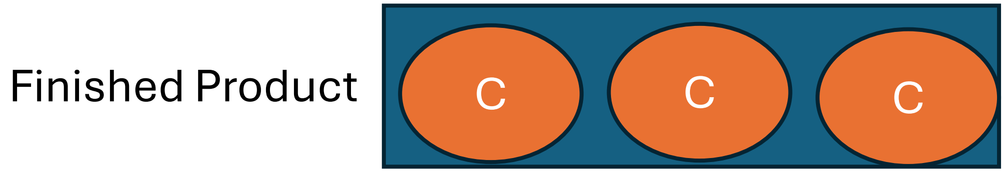
# Sequential vs simultaneous optimisation

In sequential optimisation each party optimises for its own.
In simultaneous optimisation parties synchronize with each other to find the best ‘system plan’.
Simultaneous optimisation is particularly critical when there are supply constraints.

# Simple illustration

- 2 independent companies A and B in business together.
- Each company has defined setup costs, holding costs, lead time and capacity.
- Demand for the end item is given. The end item needs 3 similar components from company B
- 3 ways to manage the situation:
  - No optimisation for any party
  - Sequential optimisation: A is optimised but not B
  - Simultaneous optimisation: the system is optimised



# Simple MRP logic

| | | | | | |
|---|---|---|---|---|---|
| Total System Cost | $10,500 | End: | $ 7,000 | Comp: | $ 3,500 |

## End Item Manufacturer

| | | | | |
|---|---|---|---|---|
| Lead Time | 2 | months | $ 7,000 | Set Up Cost |
| Setup | $ 1,000 | $/order | $ - | Holding Cost |
| Holding | $ 5.00 | $/month | $ 7,000 | Total |
| Order Capacity | 1000 | units | | |

Leadtime between Due In and POR

| Period | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gross Reqt | 0 | 0 | 0 | 0 | 0 | 30 | 50 | 200 | 30 | 70 | 180 | 40 |
| Beg Inv | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Due In | 0 | 0 | 0 | 0 | 0 | 30 | 50 | 200 | 30 | 70 | 180 | 40 |
| End Inv | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Planned Order Release | 0 | 0 | 0 | 30 | 50 | 200 | 30 | 70 | 180 | 40 | 0 | 0 |

## Tier 1 Component Manufacturer

| | | | | |
|---|---|---|---|---|
| Lead Time | 2 | months | | |
| Quantity / Assembly | 3 | units/end Item | $ 3,500 | Set Up Cost |
| Setup Cost | $ 500 | $/order | $ - | Holding Cost |
| Holding Cost | $ 1.00 | $/month | $ 3,500 | Total |
| Order Capacity | 1000 | units | | |

Output from End Item Model
becomes input to Component Model

| Period | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gross Reqt | 0 | 0 | 0 | 90 | 150 | 600 | 90 | 210 | 540 | 120 | 0 | 0 |
| Beg Inv | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Due In | 0 | 0 | 0 | 90 | 150 | 600 | 90 | 210 | 540 | 120 | 0 | 0 |
| End Inv | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Planned Order Release | 0 | 90 | 150 | 600 | 90 | 210 | 540 | 120 | 0 | 0 | 0 | 0 |

Courtesy of MIT CTL.SC2x - Supply Chain Design

# Sequential optimisation with capacity constraints at company B

A optimises its plan based on one of the method we studied in class. B tries to follow, considering its capacity limitation.

| | | | |
|---|---|---|---|
| Total System Cost | $9,100 | | |
| End: | $ 4,300 | Comp: | $ 4,800 |

**End Item Manufacturer**

| | | | | |
|---|---|---|---|---|
| Lead Time | 2 | months | $ 3,000 | Set Up Cost |
| Setup | $ 1,000 | $/order | $ 1,300 | Holding Cost |
| Holding | $ 5.00 | $/month | $ 4,300 | Total |
| Capacity | 1000 | units | | |

| Period | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gross Reqt | 0 | 0 | 0 | 0 | 0 | 30 | 50 | 200 | 30 | 70 | 180 | 40 |
| Beg Inv | 0 | 0 | 0 | 0 | 0 | 0 | 50 | 0 | 100 | 70 | 0 | 40 |
| Due In | 0 | 0 | 0 | 0 | 0 | 80 | 0 | 300 | 0 | 0 | 220 | 0 |
| Capacity | 0 | 0 | 0 | 0 | 0 | 1000 | 0 | 1000 | 0 | 0 | 1000 | 0 |
| End Inv | 0 | 0 | 0 | 0 | 0 | 50 | 0 | 100 | 70 | 0 | 40 | 0 |
| Planned Order Release | 0 | 0 | 0 | 80 | 0 | 300 | 0 | 0 | 220 | 0 | 0 | 0 |

**Tier 1 Component Manufacturer**

| | | | | |
|---|---|---|---|---|
| Lead Time | 2 | months | | |
| Quantity / Assembly | 3 | units/end Item | $ 3,000 | Set Up Cost |
| Setup Cost | $ 500 | $/order | $ 1,800 | Holding Cost |
| Holding Cost | $ 1.00 | $/month | $ 4,800 | Total |
| Capacity | 300 | units | | |

| Period | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gross Reqt | 0 | 0 | 0 | 240 | 0 | 900 | 0 | 0 | 660 | 0 | 0 | 0 |
| Beg Inv | 0 | 0 | 0 | 300 | 360 | 660 | 60 | 60 | 360 | 0 | 0 | 0 |
| Due In | 0 | 0 | 300 | 300 | 300 | 300 | 0 | 300 | 300 | 0 | 0 | 0 |
| Capacity | 0 | 0 | 300 | 300 | 300 | 300 | 0 | 300 | 300 | 0 | 0 | 0 |
| End Inv | 0 | 0 | 300 | 360 | 660 | 60 | 60 | 360 | 0 | 0 | 0 | 0 |
| Planned Order Release | 300 | 300 | 300 | 300 | 0 | 300 | 300 | 0 | 0 | 0 | 0 | 0 |

Component Mfg is constrained at 300 units.

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Courtesy of MIT CTL.SC2x - Supply Chain Design

# Simultaneous optimisation with capacity constraints at firm B

A and B are optimised jointly, using MILP method.

| Total System Cost | $8,700 |
|---|---|

| End: | $ 4,800 | Comp: | $ 3,900 |
|---|---|---|---|

**End Item Manufacturer**

| | | | | |
|---|---|---|---|---|
| Lead Time | 2 | months | $ 4,000 | Set Up Cost |
| Setup | $ 1,000 | $/order | $ 800 | Holding Cost |
| Holding | $ 5.00 | $/month | $ 4,800 | Total |
| Capacity | 1000 | units | | |

| Period | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gross Reqt | 0 | 0 | 0 | 0 | 0 | 30 | 50 | 200 | 30 | 70 | 180 | 40 |
| Beg Inv | 0 | 0 | 0 | 0 | 0 | 0 | 50 | 0 | 0 | 70 | 0 | 40 |
| Due In | 0 | 0 | 0 | 0 | 0 | 80 | 0 | 200 | 100 | 0 | 220 | 0 |
| Capacity | 0 | 0 | 0 | 0 | 0 | 1000 | 0 | 1000 | 1000 | 0 | 1000 | 0 |
| End Inv | 0 | 0 | 0 | 0 | 0 | 50 | 0 | 0 | 70 | 0 | 40 | 0 |
| Planned Order Release | 0 | 0 | 0 | 80 | 0 | 200 | 100 | 0 | 220 | 0 | 0 | 0 |

**Tier 1 Component Manufacturer**

| | | | | |
|---|---|---|---|---|
| Lead Time | 2 | months | | |
| Quantity / Assembly | 3 | units/end Item | $ 3,000 | Set Up Cost |
| Setup Cost | $ 500 | $/order | $ 900 | Holding Cost |
| Holding Cost | $ 1.00 | $/month | $ 3,900 | Total |
| Capacity | 300 | units | | |

| Period | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gross Reqt | 0 | 0 | 0 | 240 | 0 | 600 | 300 | 0 | 660 | 0 | 0 | 0 |
| Beg Inv | 0 | 0 | 0 | 0 | 60 | 360 | 60 | 60 | 360 | 0 | 0 | 0 |
| Due In | 0 | 0 | 0 | 300 | 300 | 300 | 300 | 300 | 300 | 0 | 0 | 0 |
| Capacity | 0 | 0 | 0 | 300 | 300 | 300 | 300 | 300 | 300 | 0 | 0 | 0 |
| End Inv | 0 | 0 | 0 | 60 | 360 | 60 | 60 | 360 | 0 | 0 | 0 | 0 |
| Planned Order Release | 0 | 300 | 300 | 300 | 300 | 300 | 300 | 0 | 0 | 0 | 0 | 0 |

Component Mfg is constrained at 300 units.

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Courtesy of MIT CTL.SC2x - Supply Chain Design

# Summary

- Simultaneous optimisation shows the better total cost but benefits the most to company B.
- It is difficult to implement for a complex BOM.
- It is difficult to implement without a solution to share the benefits.
- Some form of simultaneous optimisation required for inter-company transaction

| Approach | Company A cost | Company B cost |
|---|---|---|
| Simple MRP | 7000 | 4340 |
| Sequential MRP | 4300 | 4800 |
| Simultaneous MRP | 4800 | 3900 |

# Wrap-up

- MRP converts the MPS into production orders and suppliers PO considering the gross requirements, the inventory and the on order materials.
- A well run MRP will optimise cost by minimising inventory and PO exposure to suppliers
- Robust master data are an enabler to a smoot MRP

# THANK YOU

TRAILBLAZING A BETTER WORLD BY DESIGN.

SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN